

- 31.** (a) (i) What is an electric dipole ? Derive an expression for the torque acting on an electric dipole in a uniform electric field.
- (ii) An electric dipole with dipole moment 6×10^{-9} C-m is aligned at an angle of 30° with the direction of a uniform electric field of magnitude 4×10^4 NC⁻¹. Calculate magnitude of the torque acting on the electric dipole.

(i) Electric dipole-definition	1
Derivation of torque	2
(ii) Calculation of torque	2

(i) Electric Dipole:- An arrangement of two equal and opposite charges separated by a small distance.

Consider an electric dipole of charges $\pm q$ and length 2ℓ kept in a uniform electric field E . The dipole makes an angle θ with the uniform electric field. Forces on two charges due to uniform electric field, qE and $-qE$ are in opposite directions. The distance between the line of action of two force is $2l \sin \theta$. Torque due to the couple

$\tau = \text{force} \times \text{perpendicular distance between the two force}$

$$\tau = qE(2l \sin \theta)$$

$$\tau = q(2l).E \sin \theta = pE \sin \theta$$

(ii)

$$\tau = pE \sin \theta$$

$$\tau = 6 \times 10^{-9} \times 4 \times 10^4 \times \sin 30^\circ$$

$$\tau = 1.2 \times 10^{-4} \text{ Nm}$$

1

$\frac{1}{2}$

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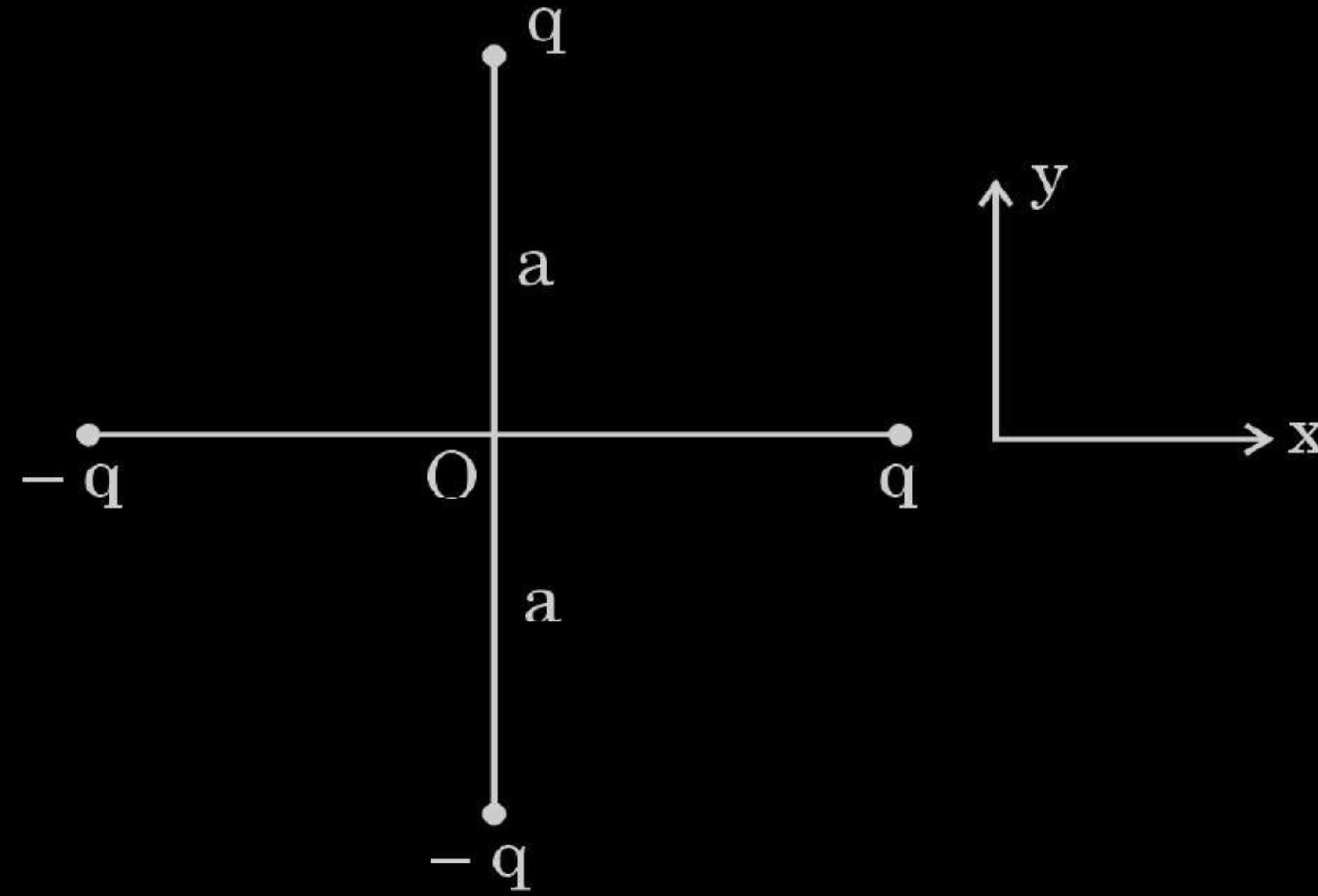
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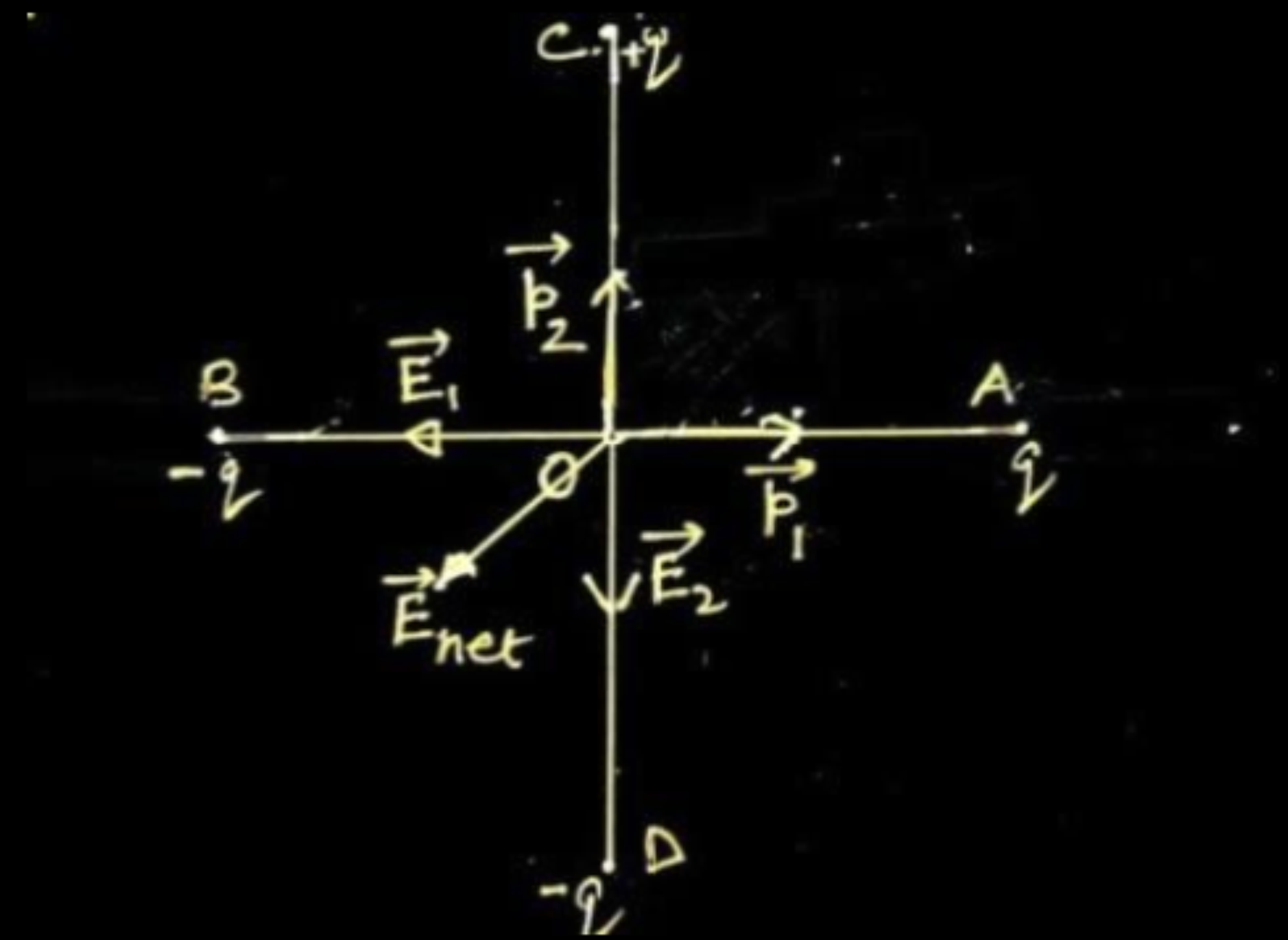
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- 19.** Two identical dipoles are arranged in x-y plane as shown in the figure. Find the magnitude and the direction of net electric field at the origin O. 2



Finding the magnitude of electric field
Finding direction of net Electric field

1 1/2
1/2



Dipole moment due to dipole BA is $\vec{p}_1$ & dipole moment due to dipole DC is $\vec{p}_2$.

Electric field $\vec{E}_1$ due to $\vec{p}_1$ is along OB.

Electric field $\vec{E}_2$ due to $\vec{p}_2$ is along OD.

Magnitude of resultant Electric field

$$|\vec{E}_{net}| = 2\sqrt{2}E$$

$$\text{Since } |\vec{E}_1| = |\vec{E}_2| = E = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{a^2}$$

$$\vec{E}_{net} = \frac{1}{4\pi\epsilon_0} \cdot \frac{2\sqrt{2}q}{a^2}$$

Direction of $\vec{E}_{net}$ is 225° to x-axis.

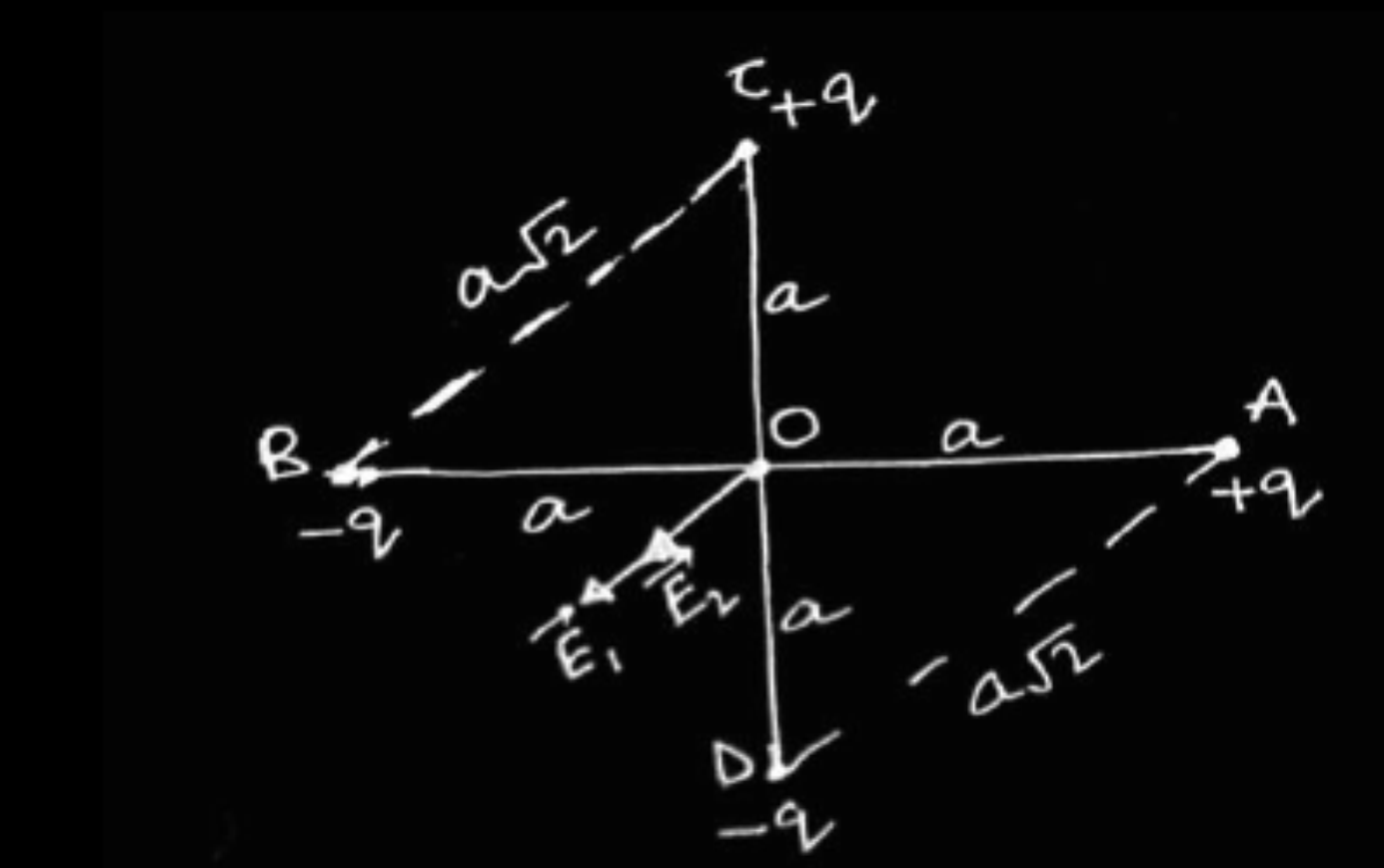
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Alternatively:



Considering CB as dipole, electric field at O

$$E_1 = \frac{2kq \times (a/\sqrt{2})}{\left[\left(\frac{a}{\sqrt{2}}\right)^2 + \left(\frac{a}{\sqrt{2}}\right)^2\right]^{3/2}}$$

$$E_1 = \frac{\sqrt{2}kq \times a}{a^3 \left(\frac{1}{2} + \frac{1}{2}\right)^{3/2}}$$

$$E_1 = \frac{\sqrt{2}kq}{a^2}$$

Similarly considering AD as another dipole, electric field at O

$$E_2 = \frac{2kq \times (a/\sqrt{2})}{\left[\left(\frac{a}{\sqrt{2}}\right)^2 + \left(\frac{a}{\sqrt{2}}\right)^2\right]^{3/2}}$$

$$E_2 = \frac{\sqrt{2}kq \times a}{a^3 \left(\frac{1}{2} + \frac{1}{2}\right)^{3/2}}$$

$$E_2 = \frac{\sqrt{2}kq}{a^2}$$

$$E_{net} = E_1 + E_2$$

$$= \frac{\sqrt{2}kq}{a^2} + \frac{\sqrt{2}kq}{a^2}$$

$$= \frac{2\sqrt{2}kq}{a^2}$$

Direction of $\vec{E}_{net}$ is 225° to x-axis.

1/2

1/2

1/2

1/2

9. A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be :

(a) $\frac{\vec{F}}{16}$

(b) $\frac{\vec{F}}{8}$

(c) $\frac{\vec{F}}{4}$

(d) $\frac{\vec{F}}{2}$

9. A point charge situated at a distance 'r' from a short electric dipole on its axis, experiences a force $\vec{F}$. If the distance of the charge is '2r', the force on the charge will be :

(a) $\frac{\vec{F}}{16}$

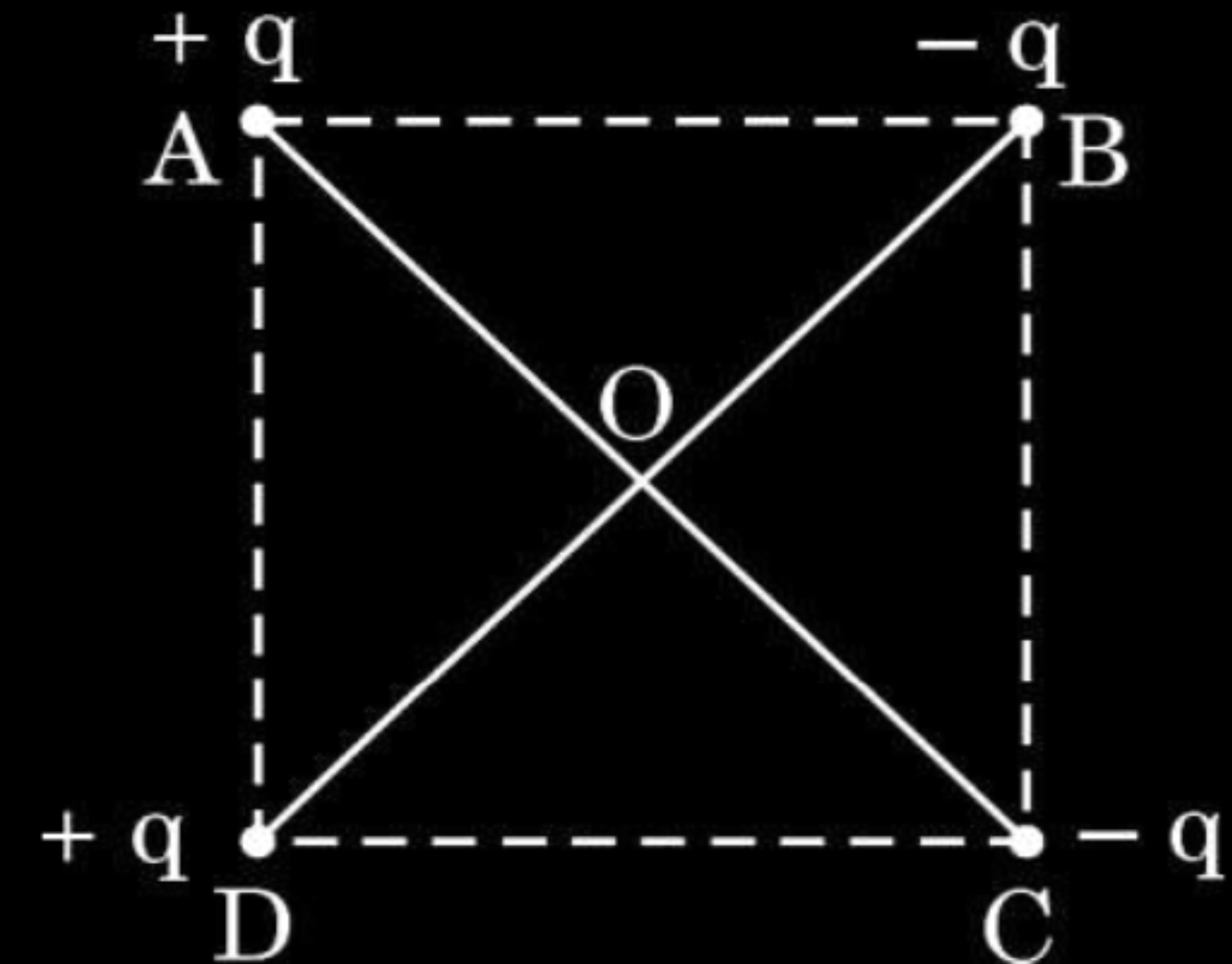
(b) $\frac{\vec{F}}{8}$

(c) $\frac{\vec{F}}{4}$

(d) $\frac{\vec{F}}{2}$

(b) $\frac{\vec{F}}{8}$

- (b) (i) Derive an expression for the electric field at a point on the equatorial plane of an electric dipole consisting of charges q and $-q$ separated by a distance $2a$.
- (ii) The distance of a far off point on the equatorial plane of an electric dipole is halved. How will the electric field be affected for the dipole ?
- (iii) Two identical electric dipoles are placed along the diagonals of a square ABCD of side $\sqrt{2}$ m as shown in the figure. Obtain the magnitude and direction of the net electric field at the centre (O) of the square.



i) Derivation of electric field	2
ii) Effect on electric field	1
iii) Finding magnitude and direction of electric field	2

$$i) \quad E_{+q} = \frac{q}{4\pi \epsilon_0} \times \frac{1}{r^2 + a^2}$$

$$E_{-q} = \frac{q}{4\pi \epsilon_0} \times \frac{1}{r^2 + a^2}$$

The components normal to dipole axis cancel away. The components along the dipole axis add up.

Total electric field is opposite to dipole moment.

$$\begin{aligned} \vec{E} &= -(E_{+q} + E_{-q}) \cos \theta \hat{p} \\ &= \frac{-2qa}{4\pi \epsilon_0 (r^2 + a^2)^{3/2}} \hat{p} \\ &= \frac{-\vec{p}}{4\pi \epsilon_0 (r^2 + a^2)^{3/2}} \end{aligned}$$

Deduct 1/2 mark if the expression of electric field is not in vector form.

ii) At far off point $r \gg a$

$$\vec{E} = \frac{-\vec{p}}{4\pi \epsilon_0 r^3}$$

When distance is halved.

$$\vec{E} = \frac{-\vec{p}}{4\pi \epsilon_0 \left(\frac{r}{2}\right)^3}$$

$$= \frac{-8\vec{p}}{4\pi \epsilon_0 r^3}$$

$\vec{E}$ becomes 8 times

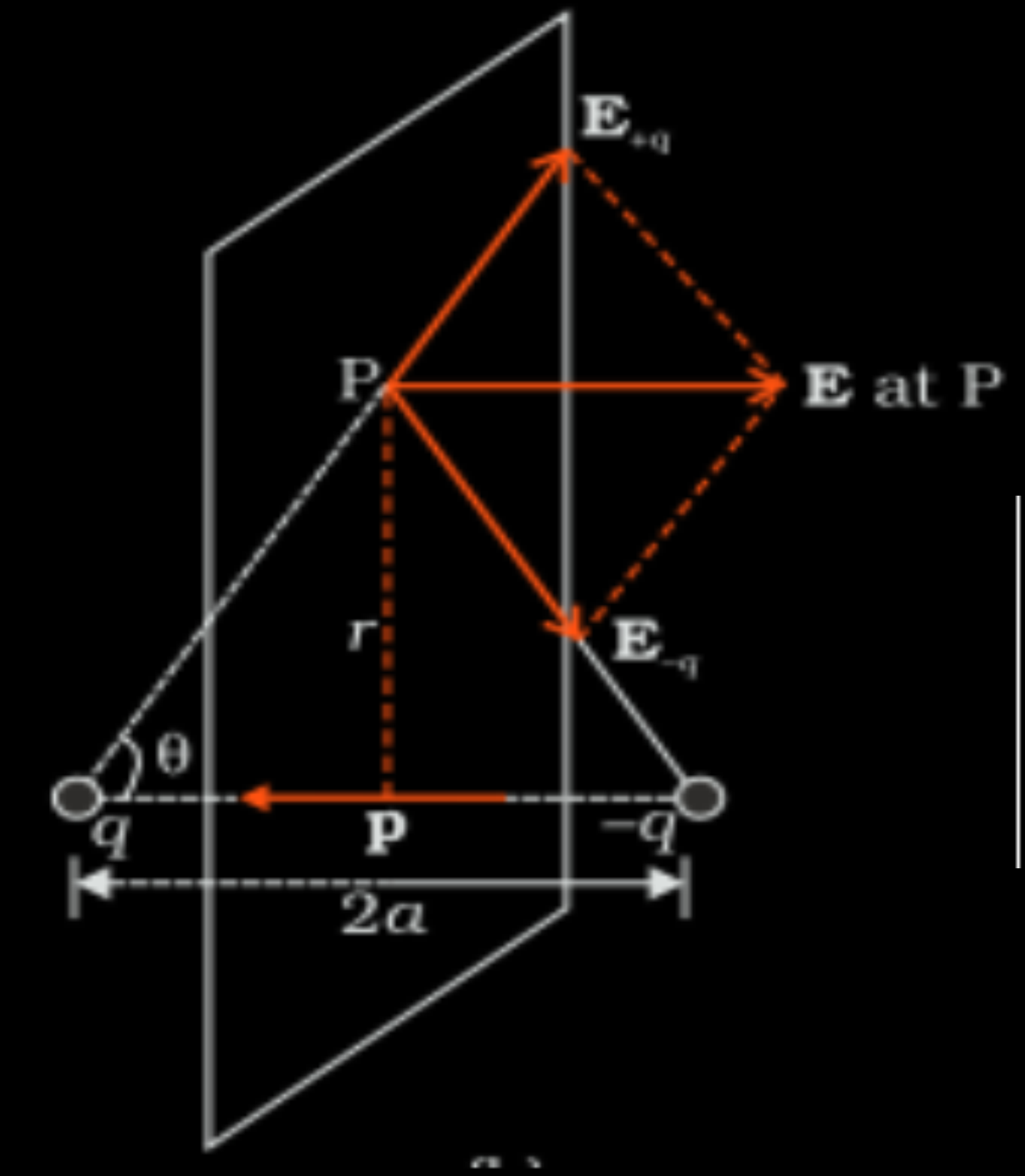
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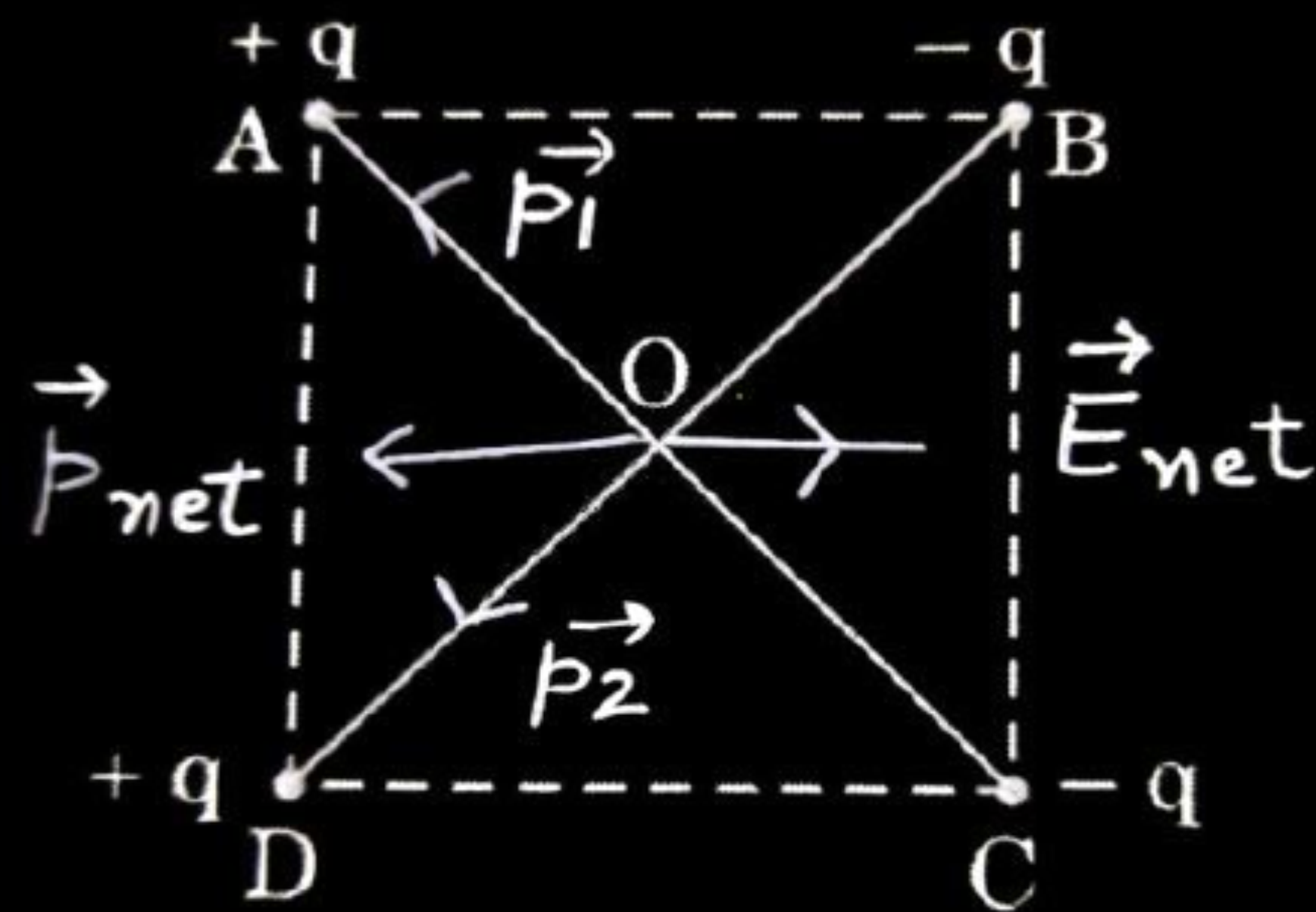
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iii)



$$p_1 = q \times 2Cm \quad (\text{along } OA)$$

$$p_2 = q \times 2Cm \quad (\text{along } OD)$$

$$p_{net} = \sqrt{p_1^2 + p_2^2}$$

$$= 2\sqrt{2} q Cm$$

Electric field at centre O

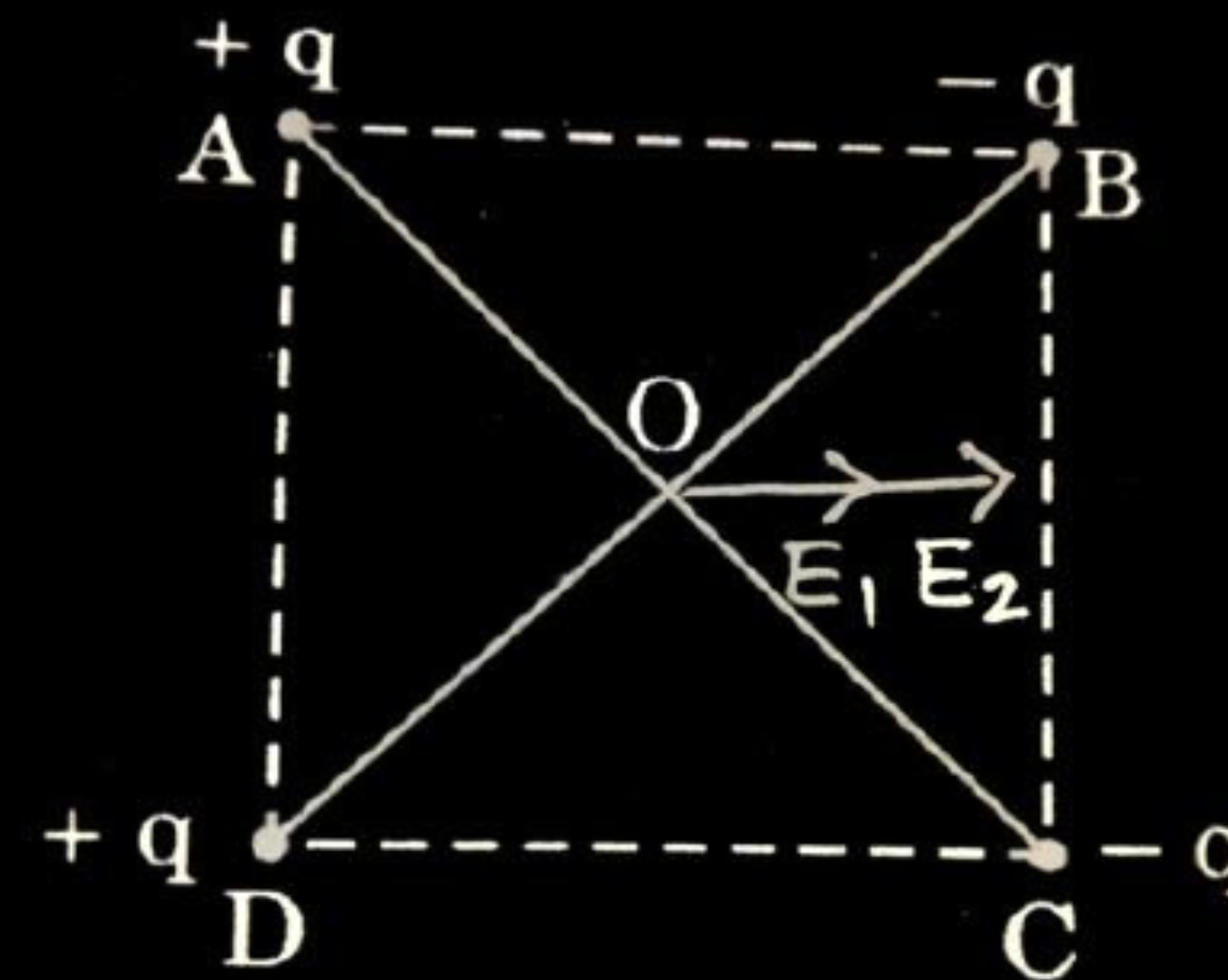
$$E = \frac{k p_{net}}{(r^2 + a^2)^{3/2}}$$

at point O, $r = 0$, $a = 1 \text{ m}$

$$E = \frac{k \times 2\sqrt{2}q}{1^3} = 2\sqrt{2}kq = \frac{2\sqrt{2}q}{4\pi \epsilon_o}$$

Along DC

Alternatively



Considering AB as dipole, electric field at O

$$E_1 = \frac{2kq \times a}{((\frac{1}{\sqrt{2}})^2 + (\frac{1}{\sqrt{2}})^2)^{3/2}} = \frac{2kqa}{(\frac{1}{2} + \frac{1}{2})^{3/2}} = 2kqa$$

Similarly considering DC as another dipole, electric field at O

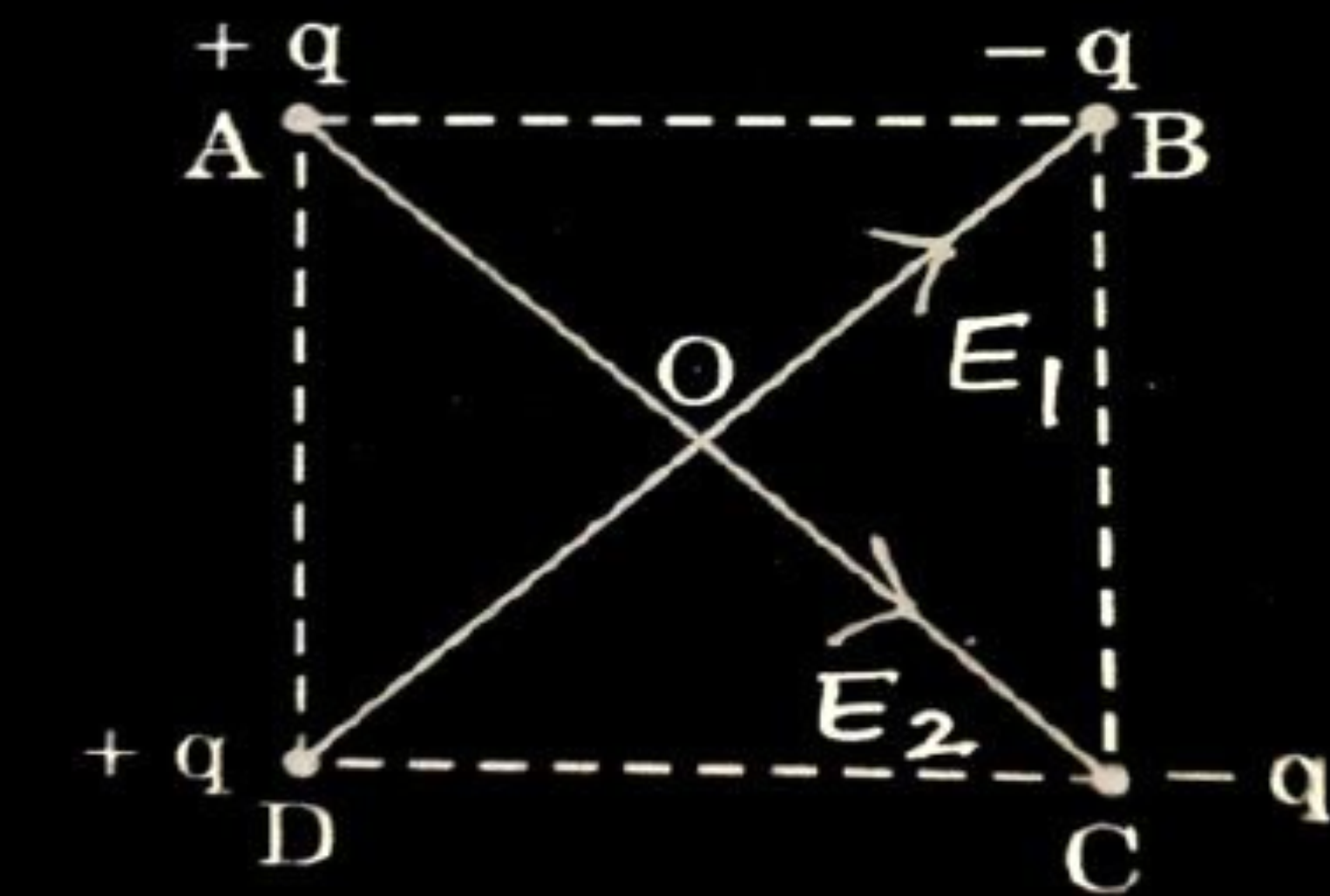
$$E_2 = \frac{2kq \times a}{((\frac{1}{\sqrt{2}})^2 + (\frac{1}{\sqrt{2}})^2)^{3/2}} = \frac{2kqa}{(\frac{1}{2} + \frac{1}{2})^{3/2}} = 2kqa$$

$$E_{net} = E_1 + E_2 = 4kqa = \frac{1}{4\pi \epsilon_o} \times 4 \times \frac{1}{\sqrt{2}} \times q$$

$$= 2\sqrt{2}kq = \frac{2\sqrt{2}q}{4\pi \epsilon_o}$$

Along DC

Alternatively



$$E = \frac{kq}{r^2}$$

$$AC = BD = 2m$$

$$r = OA = OB = OC = OD = 1m$$

Electric field at O due to charges at B and D

$$E_1 = E_B + E_D$$

$$E_1 = \frac{kq}{1^2} + \frac{kq}{1^2} \quad \text{along } OB$$

$$= 2kq$$

Electric field at O due to charges at A and C

$$E_2 = E_A + E_C$$

$$E_2 = \frac{kq}{1^2} + \frac{kq}{1^2}$$

$$= 2kq \quad \text{along } OC$$

$$E_{net} = \sqrt{E_1^2 + E_2^2}$$

$$= 2\sqrt{2}kq = \frac{2\sqrt{2}q}{4\pi \epsilon_o}$$

Along DC

7. An electric dipole of dipole moment 2×10^{-8} C-m in a uniform electric field experiences a maximum torque of 6×10^{-4} N-m. The magnitude of electric field is

(a) $2.2 \times 10^3 \text{ Vm}^{-1}$

(b) $1.2 \times 10^4 \text{ Vm}^{-1}$

(c) $3.0 \times 10^4 \text{ Vm}^{-1}$

(d) $4.2 \times 10^3 \text{ Vm}^{-1}$

7. An electric dipole of dipole moment 2×10^{-8} C-m in a uniform electric field experiences a maximum torque of 6×10^{-4} N-m. The magnitude of electric field is

1

(a) $2.2 \times 10^3 \text{ Vm}^{-1}$

(b) $1.2 \times 10^4 \text{ Vm}^{-1}$

(c) $3.0 \times 10^4 \text{ Vm}^{-1}$

(d) $4.2 \times 10^3 \text{ Vm}^{-1}$

(c) $3 \times 10^4 \text{ Vm}^{-1}$

1. An electric dipole of length 2 cm is placed at an angle of 30° with an electric field 2×10^5 N/C. If the dipole experiences a torque of 8×10^{-3} Nm, the magnitude of either charge of the dipole, is

(A) $4 \mu\text{C}$

(B) $7 \mu\text{C}$

(C) 8 mC

(D) 2 mC

1. An electric dipole of length 2 cm is placed at an angle of 30° with an electric field 2×10^5 N/C. If the dipole experiences a torque of 8×10^{-3} Nm, the magnitude of either charge of the dipole, is

(A) $4 \mu\text{C}$

(B) $7 \mu\text{C}$

(C) 8 mC

(D) 2 mC

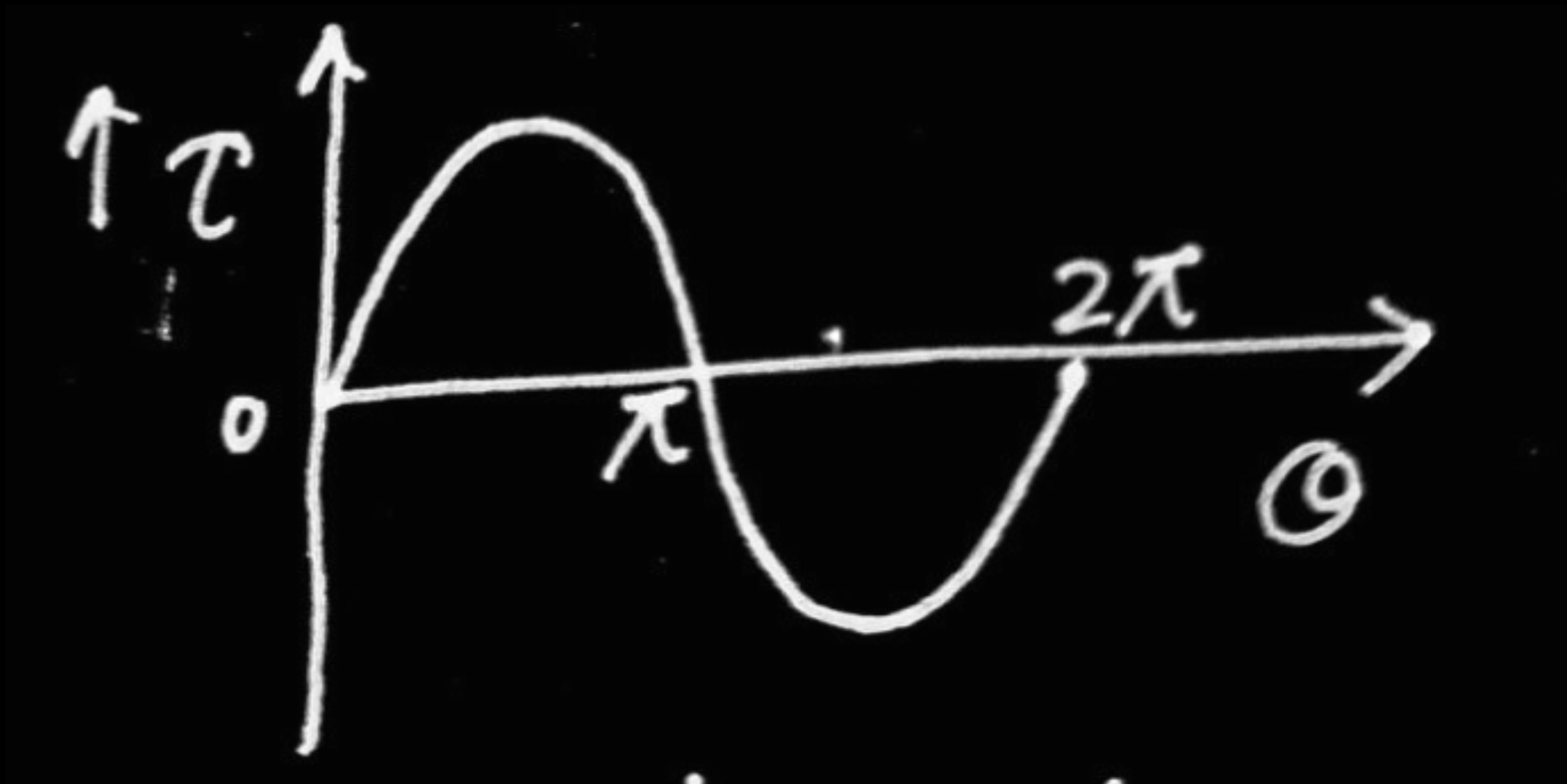
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(A) $4\mu\text{C}$

22. An electric dipole of dipole moment ($\vec{p}$) is kept in a uniform electric field $\vec{E}$. Show graphically the variation of torque acting on the dipole (τ) with its orientation (θ) in the field. Find the orientation in which torque is (i) zero and (ii) maximum.

2

- | | |
|---|-------------------|
| (a) Graphical representation of torque - θ | (1) |
| Finding the orientation for | |
| (i) Zero torque | ($\frac{1}{2}$) |
| (ii) Maximum torque | ($\frac{1}{2}$) |



1

(i) Torque is zero for orientation corresponding to $\theta = 0^\circ$ and $\theta = 180^\circ$

$\frac{1}{2}$

(ii) Torque is maximum for orientation corresponding to $\theta = \frac{\pi}{2}$ and $\frac{3\pi}{2}$

$\frac{1}{2}$

(Note: Award full credit of 1 mark for parts (i) and (ii) if a student writes only one value of angle in each case.)

1. An electric dipole with dipole moment $\vec{P} = P_0 \hat{i} - P_0 \hat{j}$ is placed in an electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$, where P_0 , E_1 and E_2 are constants. The torque $\vec{\tau}$ acting on the dipole is :

(a) $P_0 (E_2 - E_1) \hat{k}$

(b) $P_0 (E_2 + E_1) \hat{k}$

(c) $-P_0 (E_2 + E_1) \hat{k}$

(d) $P_0 (E_1 - E_2) \hat{k}$

1. An electric dipole with dipole moment $\vec{P} = P_0 \hat{i} - P_0 \hat{j}$ is placed in an electric field $\vec{E} = E_1 \hat{i} + E_2 \hat{j}$, where P_0 , E_1 and E_2 are constants. The torque $\vec{\tau}$ acting on the dipole is :

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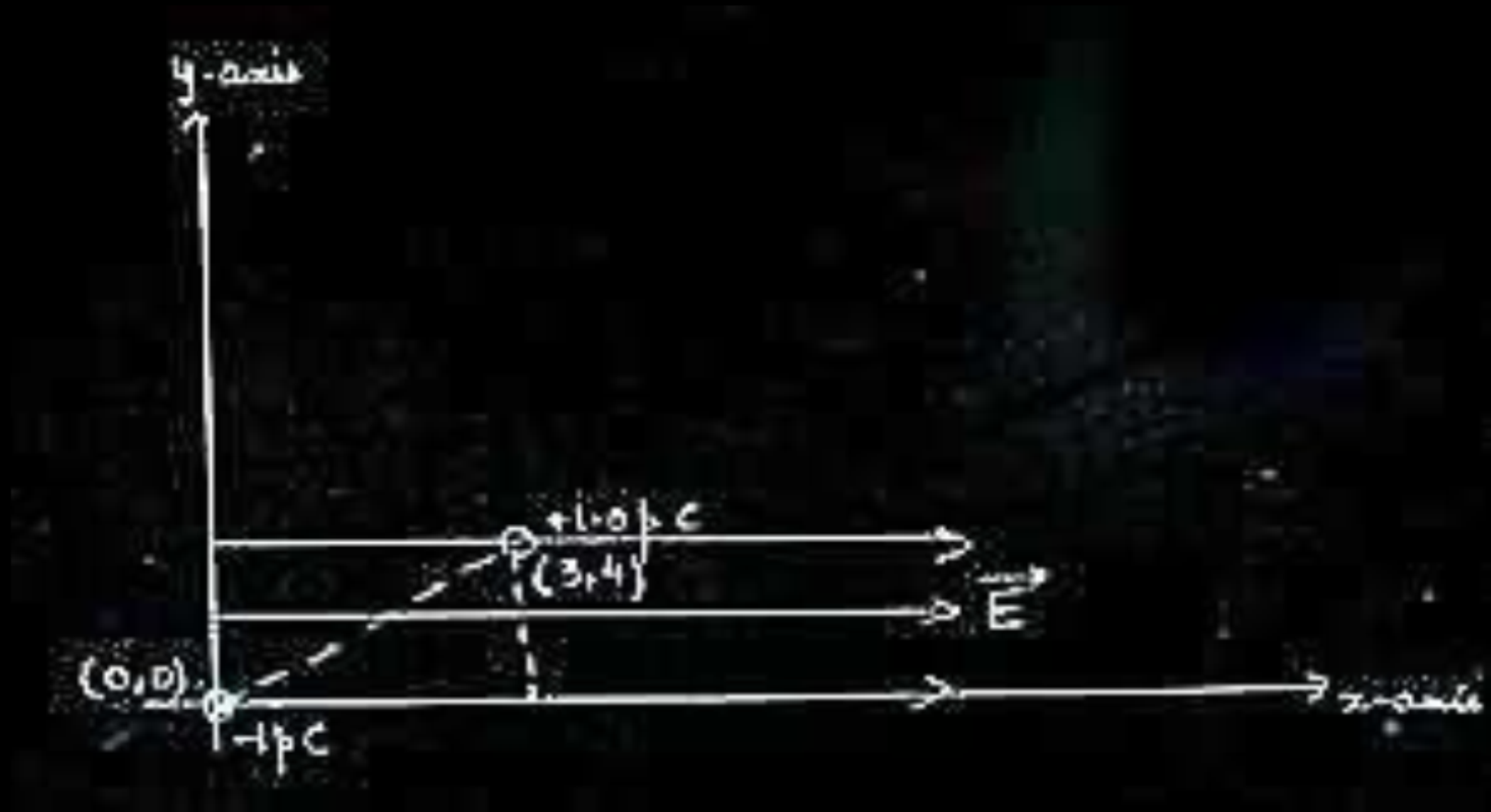
(c) $-P_0 (E_2 + E_1) \hat{k}$

(d) $P_0 (E_1 - E_2) \hat{k}$

(b) $p_0(E_2 + E_1) k$

- (ii) An electric dipole consists of point charges -1.0 pC and $+1.0 \text{ pC}$ located at $(0, 0)$ and $(3 \text{ mm}, 4 \text{ mm})$ respectively in $x - y$ plane. An electric field $\vec{E} = \left(\frac{1000 \text{ V}}{\text{m}} \right) \hat{i}$ is switched on in the region. Find the torque $\vec{\tau}$ acting on the dipole.

(ii)



$$\tau = pE \sin \theta$$

$$= (2aq) E \sin \theta$$

$$= (5 \times 10^{-3} \times 1 \times 10^{-12}) 10^3 \times \frac{4}{5}$$

$$= 4 \times 10^{-12} \text{ Nm}$$

Direction is along -ve Z direction.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- 30.** An electric dipole is a pair of equal and opposite point charges $+q$ and $-q$ separated by a distance $2a$. A dipole can be characterised by dipole moment determined by its charge and separation between the two charges. It is a vector quantity. By convention the direction of a dipole is taken to be the direction from $-q$ to $+q$. Like fields due to a point charge, fields due to a given dipole at a point can be found.

When a dipole is placed in a uniform external electric field, it experiences a torque. This torque tends to rotate the dipole and work is done in this process. This work is stored as the potential energy of the system.

- (i) An electric dipole is kept in a uniform electric field such that the dipole moment is not along the electric field. It will experience :

- (A) A net force and a torque
- (B) A net force but no torque
- (C) A torque but no net force
- (D) neither a net force nor a torque

- (ii) An electric dipole consists of charges $+q$ and $-q$, separated by distance $2a$. The force acting on a point charge placed at a distance x ($\gg a$) on the axis of the dipole is F . If the distance of the charge is doubled, the force will become :

1

(A) $\frac{F}{16}$

(B) $\frac{F}{8}$

(C) $\frac{F}{4}$

(D) $\frac{F}{2}$

- (iii) Two identical small electric dipoles, each of dipole moment $\vec{p}$ are arranged perpendicular to each other such that their negative charges coincide. The magnitude of the net dipole moment of the arrangement will be :

1

(A) p

(B) $2p$

(C) $p\sqrt{2}$

(D) $\frac{p}{\sqrt{2}}$

(iv) (a) A molecule with dipole moment 3.0×10^{-30} C-m is placed in a uniform electric field of 2×10^4 NC⁻¹. The maximum torque experienced by the molecule is :

1

- (A) 1.5×10^{-34} N-m
- (B) 6.0×10^{-26} N-m
- (C) 4.5×10^{34} N-m
- (D) 6.0×10^{-34} N-m

30	(i)	(C)	A torque but no net force	1
	(ii)	(B)	$F/8$	1
	(iii)	(C)	$p\sqrt{2}$	1
	(iv)	(a) (B)	$6 \times 10^{-26} \text{ Nm}$	1

- (i) Two identical electric dipoles, each consisting of charges $-q$ and $+q$ separated by distance d , are arranged in x - y plane such that their negative charges lie at the origin O and positive charges lie at points $(d, 0)$ and $(0, d)$ respectively. The net dipole moment of the system is : l

(A) $-q d (\hat{i} + \hat{j})$

(B) $q d (\hat{i} + \hat{j})$

(C) $q d (\hat{i} - \hat{j})$

(D) $q d (\hat{j} - \hat{i})$

- (i) Two identical electric dipoles, each consisting of charges $-q$ and $+q$ separated by distance d , are arranged in x-y plane such that their negative charges lie at the origin O and positive charges lie at points $(d, 0)$ and $(0, d)$ respectively. The net dipole moment of the system is :

1

(A) $-q d (\hat{i} + \hat{j})$

(B) $q d (\hat{i} + \hat{j})$

(C) $q d (\hat{i} - \hat{j})$

(D) $q d (\hat{j} - \hat{i})$

(B) $qd(\hat{i} + \hat{j})$

(ii) E_1 and E_2 are magnitudes of electric field due to a dipole, consisting of charges $-q$ and $+q$ separated by distance $2a$, at points r ($\gg a$) (1) on its axis, and (2) on equatorial plane, respectively. Then $\left(\frac{E_1}{E_2}\right)$ is : 1

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 2

(D) 4

(ii) E_1 and E_2 are magnitudes of electric field due to a dipole, consisting of charges $-q$ and $+q$ separated by distance $2a$, at points r ($\gg a$) (1) on its axis, and (2) on equatorial plane, respectively. Then $\left(\frac{E_1}{E_2}\right)$ is : 1

(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) 2

(D) 4

(C) 2

- (iii) An electric dipole of dipole moment 5.0×10^{-8} Cm is placed in a region where an electric field of magnitude 1.0×10^3 N/C acts at a given instant. At that instant the electric field $\vec{E}$ is inclined at an angle of 30° to dipole moment $\vec{P}$. The magnitude of torque acting on the dipole, at that instant is :

(A) 2.5×10^{-5} Nm

(B) 5.0×10^{-5} Nm

(C) 1.0×10^{-4} Nm

(D) 2.0×10^{-6} Nm

- (iii) An electric dipole of dipole moment $5.0 \times 10^{-8} \text{ Cm}$ is placed in a region where an electric field of magnitude $1.0 \times 10^3 \text{ N/C}$ acts at a given instant. At that instant the electric field $\vec{E}$ is inclined at an angle of 30° to dipole moment $\vec{P}$. The magnitude of torque acting on the dipole, at that instant is :

(A) $2.5 \times 10^{-5} \text{ Nm}$

(B) $5.0 \times 10^{-5} \text{ Nm}$

(C) $1.0 \times 10^{-4} \text{ Nm}$

(D) $2.0 \times 10^{-6} \text{ Nm}$

(A) $2.5 \times 10^{-5} \text{ Nm}$

- (ii) An electric dipole of dipole moment $\vec{p} = (0.8 \hat{i} + 0.6 \hat{j}) 10^{-29} \text{ Cm}$ is placed in an electric field $\vec{E} = 1.0 \times 10^7 \hat{k} \frac{\text{V}}{\text{m}}$. Calculate the magnitude of the torque acting on it and the angle it makes with the x-axis, at this instant.

$$(ii) \quad \vec{\tau} = \vec{p} \times \vec{E}$$

$$= (0.8\hat{i} + 0.6\hat{j}) \times 10^{-29} \times (1 \times 10^7) \hat{k}$$

$$= [0.8(-\hat{j}) + 0.6\hat{i}] \times 10^{-22}$$

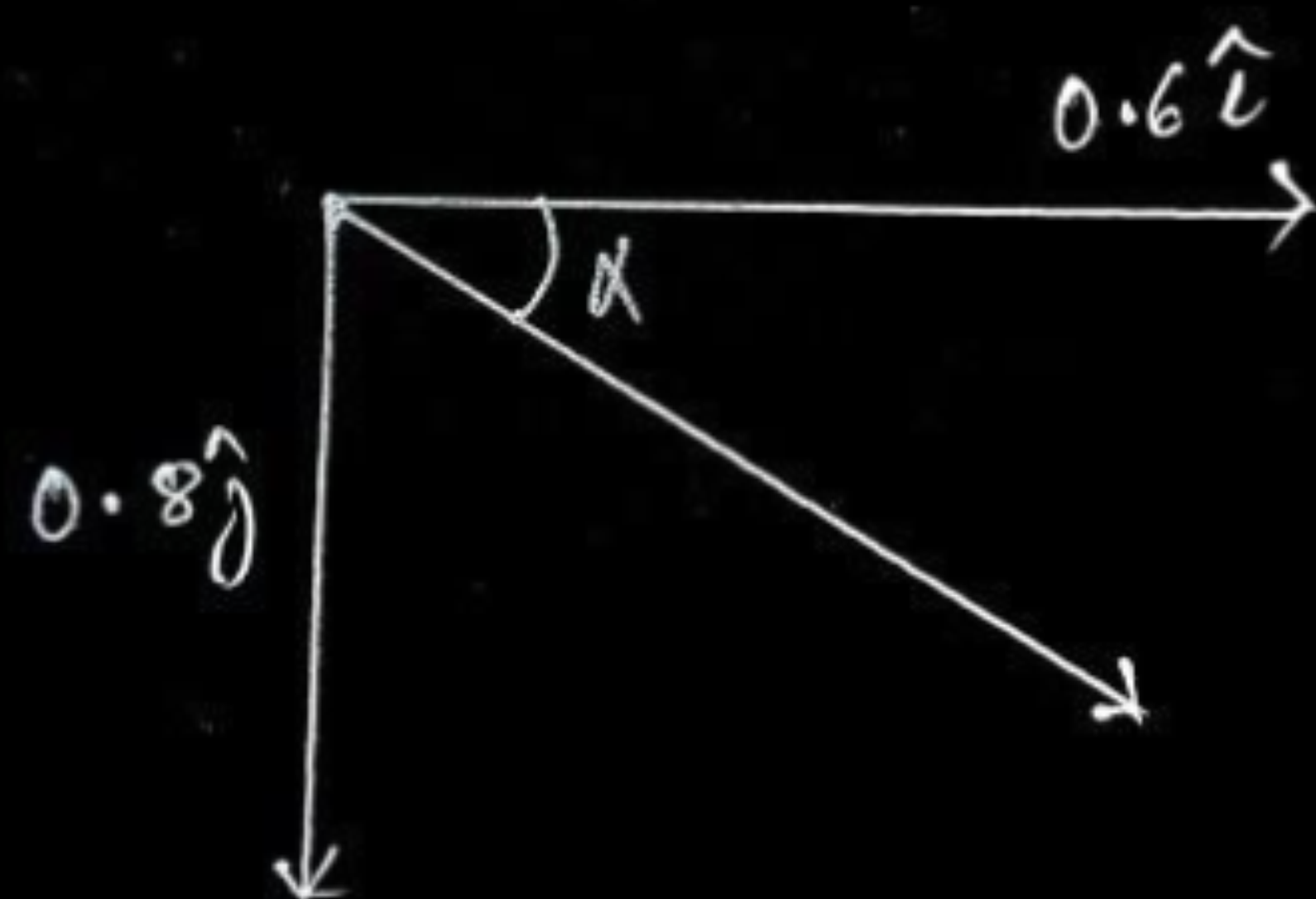
$$\tau = \left[\sqrt{(0.8)^2 + (0.6)^2} \right] \times 10^{-22}$$

$$= 10^{-22} \text{ Nm}$$

$$\tan \alpha = \frac{|0.8|}{0.6}$$

$$\alpha = \tan^{-1} \left(\frac{4}{3} \right)$$

$$\alpha = 53^\circ$$



$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (b) (i) Obtain an expression for the electric field $\vec{E}$ due to a dipole of dipole moment $\vec{p}$ at a point on its equatorial plane and specify its direction. Hence, find the value of electric field :
- (I) at the centre of the dipole ($r = 0$), and
- (II) at a point $r \gg a$,
- where $2a$ is the length of the dipole.

(i)	Obtaining expression for electric field due to a dipole on its equatorial plane	2
	Finding electric field:	
	(I) At centre of the dipole	$\frac{1}{2}$
	(II) At a point $r \gg a$	$\frac{1}{2}$
(ii)	Calculating net electric flux through cube	2

The magnitudes of the electric field due to two charges +q and -q are

$$E_{+q} = \frac{q}{4\pi\epsilon_0} \frac{1}{(r^2 + a^2)}$$

$$E_{-q} = \frac{q}{4\pi\epsilon_0} \frac{1}{(r^2 + a^2)}$$

The total electric field

$$\vec{E} = -(E_{+q} + E_{-q}) \cos \theta \hat{p}$$

$$\vec{E} = -\frac{\vec{p}}{4\pi\epsilon_0 (r^2 + a^2)^{3/2}}$$

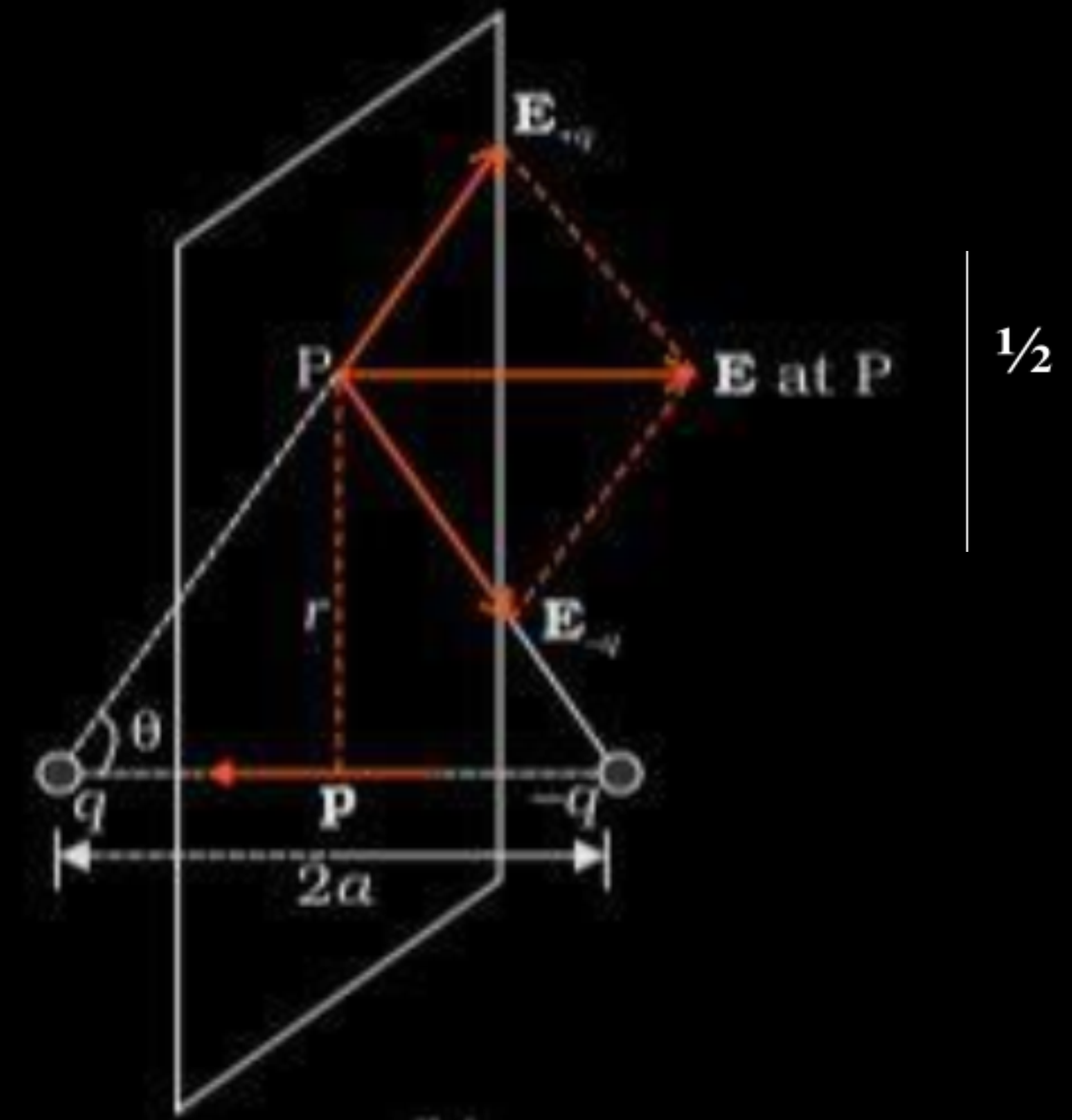
Direction of electric field is opposite to dipole moment ($\vec{p}$)

(I) At centre of dipole, $r = 0$

$$\vec{E} = -\frac{-\vec{p}}{4\pi\epsilon_0 a^3}$$

(II) At a point $r \gg a$

$$\vec{E} = -\frac{-\vec{p}}{4\pi\epsilon_0 r^3}$$



- 32.** (a) (i) Two point charges $+q$ and $-q$ are held at $(a, 0)$ and $(-a, 0)$ in x - y plane. Obtain an expression for the net electric field due to the charges at a point $(0, y)$. Hence, find electric field at a far off point ($y \gg a$).

- | | |
|---|---|
| (i) Finding electric field at a far off point ($y \gg a$) | 3 |
| (ii) Calculation of work done in shifting the charges | 2 |

Magnitude of electric field due to the two charges +q and -q are given by

$$E_{+q} = \frac{q}{4\pi\epsilon_0} \frac{1}{y^2 + a^2}$$

$$E_{-q} = \frac{q}{4\pi\epsilon_0} \frac{1}{y^2 + a^2}$$

Components normal to the dipole axis cancel out.

The components along the dipole axis add up.

The total electric field is opposite to the dipole moment will be given by-

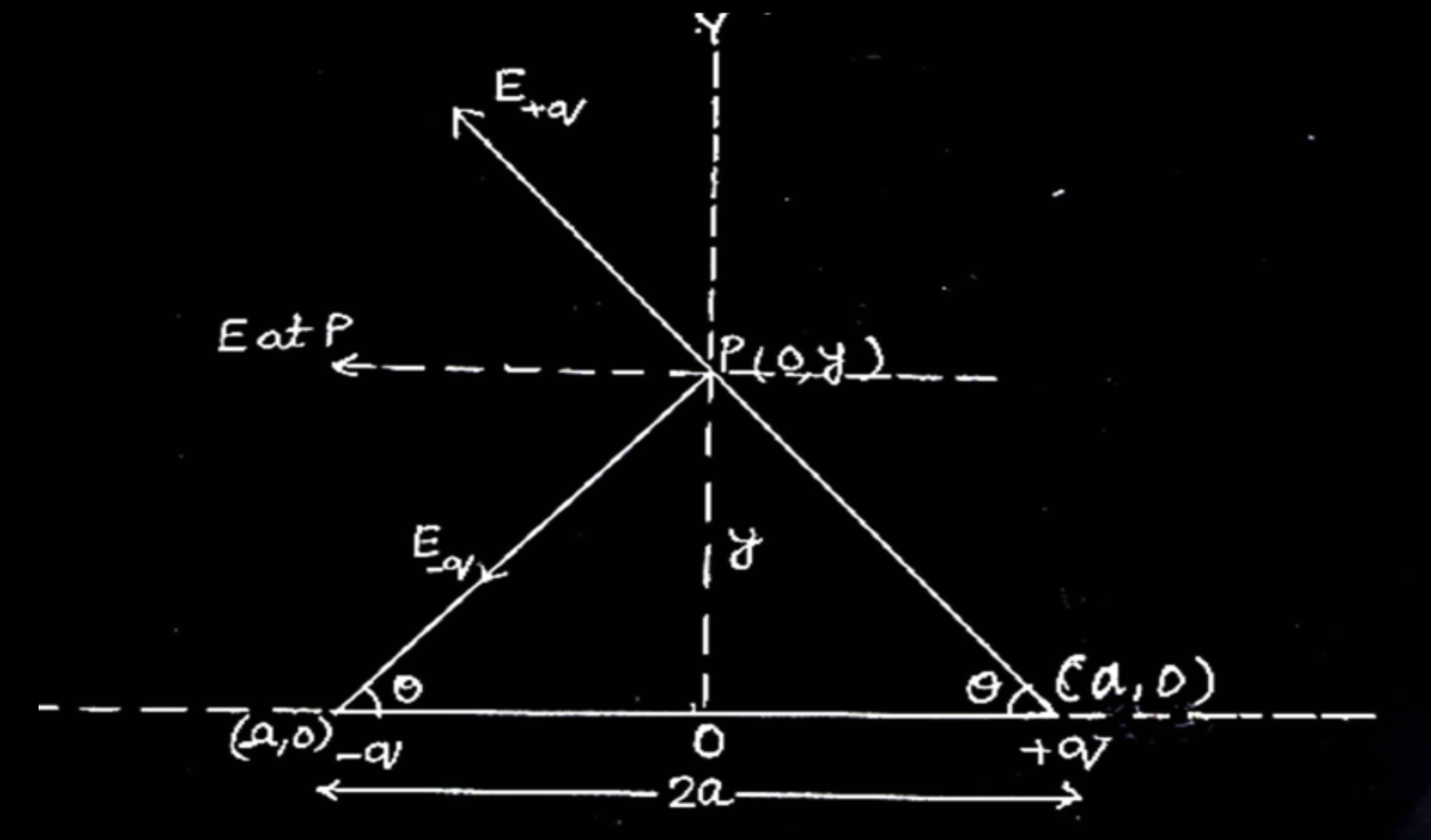
$$\begin{aligned} \vec{E} &= - (E_{+q} + E_{-q}) \cos \theta \hat{p} \\ &= - \frac{2qa}{4\pi\epsilon_0 (y^2 + a^2)^{3/2}} \hat{p} \quad (\hat{p} \text{ is a unit vector along dipole moment}) \end{aligned}$$

At large distance ($y \gg a$)

$$\vec{E} = \frac{-2qa}{4\pi\epsilon_0 y^3} \hat{p}$$

1/2

1/2



1/2

1/2

1/2

1/2

1. An electric dipole of dipole moment 1.0×10^{-12} Cm lies along x-axis. An electric field of magnitude 2.0×10^4 NC⁻¹ is switched on at an instant in the region. The unit vector along the electric field is $\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$. The magnitude of the torque acting on the dipole at that instant is :

(A) 0.5×10^{-6} Nm

(B) 1.0×10^{-8} Nm

(C) 2.0×10^{-8} Nm

(D) 4.0×10^{-8} Nm

1. An electric dipole of dipole moment 1.0×10^{-12} Cm lies along x-axis. An electric field of magnitude 2.0×10^4 NC⁻¹ is switched on at an instant in the region. The unit vector along the electric field is $\frac{\sqrt{3}}{2} \hat{i} + \frac{1}{2} \hat{j}$. The magnitude of the torque acting on the dipole at that instant is :

(A) 0.5×10^{-6} Nm

(B) 1.0×10^{-8} Nm

(C) 2.0×10^{-8} Nm

(D) 4.0×10^{-8} Nm

(B) 1.0×10^{-8} Nm

1. Consider two identical dipoles D_1 and D_2 . Charges $-q$ and q of dipole D_1 are located at $(0, 0)$ and $(a, 0)$ and that of dipole D_2 at $(0, a)$ and $(0, 2a)$ in x - y plane, respectively. The net dipole moment of the system is

1

(A) $qa(\hat{i} + \hat{j})$

(B) $-qa(\hat{i} + \hat{j})$

(C) $qa(\hat{i} - \hat{j})$

(D) $-qa(\hat{i} - \hat{j})$

1. Consider two identical dipoles D_1 and D_2 . Charges $-q$ and q of dipole D_1 are located at $(0, 0)$ and $(a, 0)$ and that of dipole D_2 at $(0, a)$ and $(0, 2a)$ in x - y plane, respectively. The net dipole moment of the system is **1**

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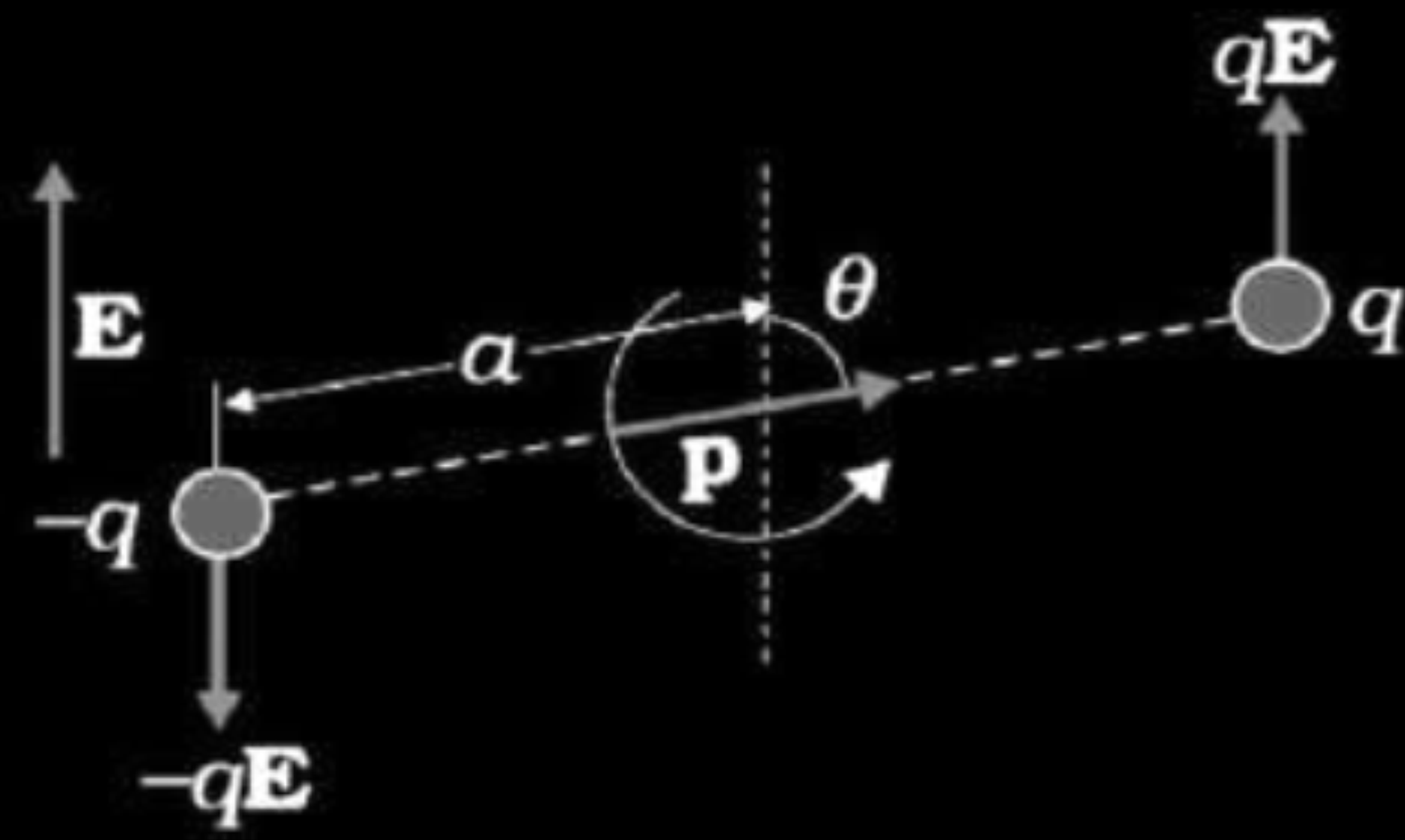
(D) $-qa(\hat{i} - \hat{j})$

(A) $qa(\hat{i} + \hat{j})$

21. Derive the expression for the torque acting on an electric dipole, when it is held in a uniform electric field. Identify the orientation of the dipole in the electric field, in which it attains a stable equilibrium.

2

Derivation of the expression for the torque	1½
Identification of the orientation of stable equilibrium	½



Force on q is $q\mathbf{E}$ and a force on $-q$ is $-q\mathbf{E}$.

Hence torque

$$\tau = qE \times 2a \sin \theta$$

$$\tau = PE \sin \theta$$

$$\vec{\tau} = \vec{P} \times \vec{E}$$

For stable equilibrium $\theta = 0^\circ$

1/2

1/2

1/2

1/2

11. Torque acting on an electric dipole placed in an electric field is maximum when the angle between the electric field and the dipole moment is _____.

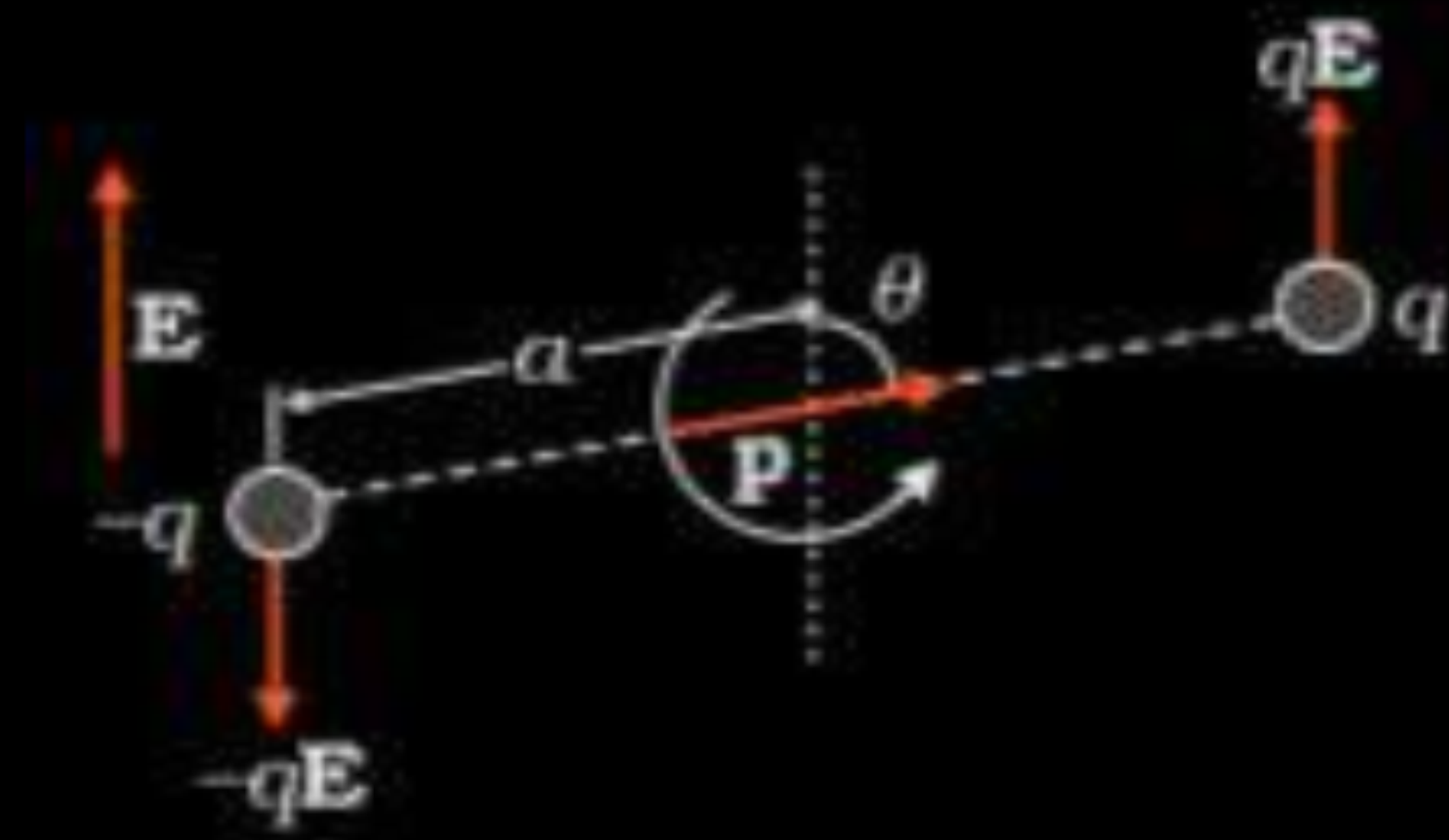
11. Torque acting on an electric dipole placed in an electric field is maximum when the angle between the electric field and the dipole moment is _____.

1

$$90^0 \text{ or } \frac{\pi}{2}$$

- 35.** (a) An electric dipole of dipole moment $\vec{p}$ is placed in a uniform electric field $\vec{E}$ at an angle θ with it. Derive the expression for torque ($\vec{\tau}$) acting on it. Find the orientation of the dipole relative to the electric field for which torque on it is (i) maximum, and (ii) half of maximum.

a)



1/2

From diagram

$$\begin{aligned}\text{Magnitude of Torque} &= (qE)(2a \sin\theta) \\ &= (2qa)(E \sin\theta) \\ &= pE \sin\theta\end{aligned}$$

1/2

1/2

For direction

$$\vec{\tau} = \vec{p} \times \vec{E}$$

1/2

i) for maximum Torque, dipole should be placed perpendicular to the direction of electric field

$$\theta = 90^\circ = \frac{\pi}{2}$$

1/2

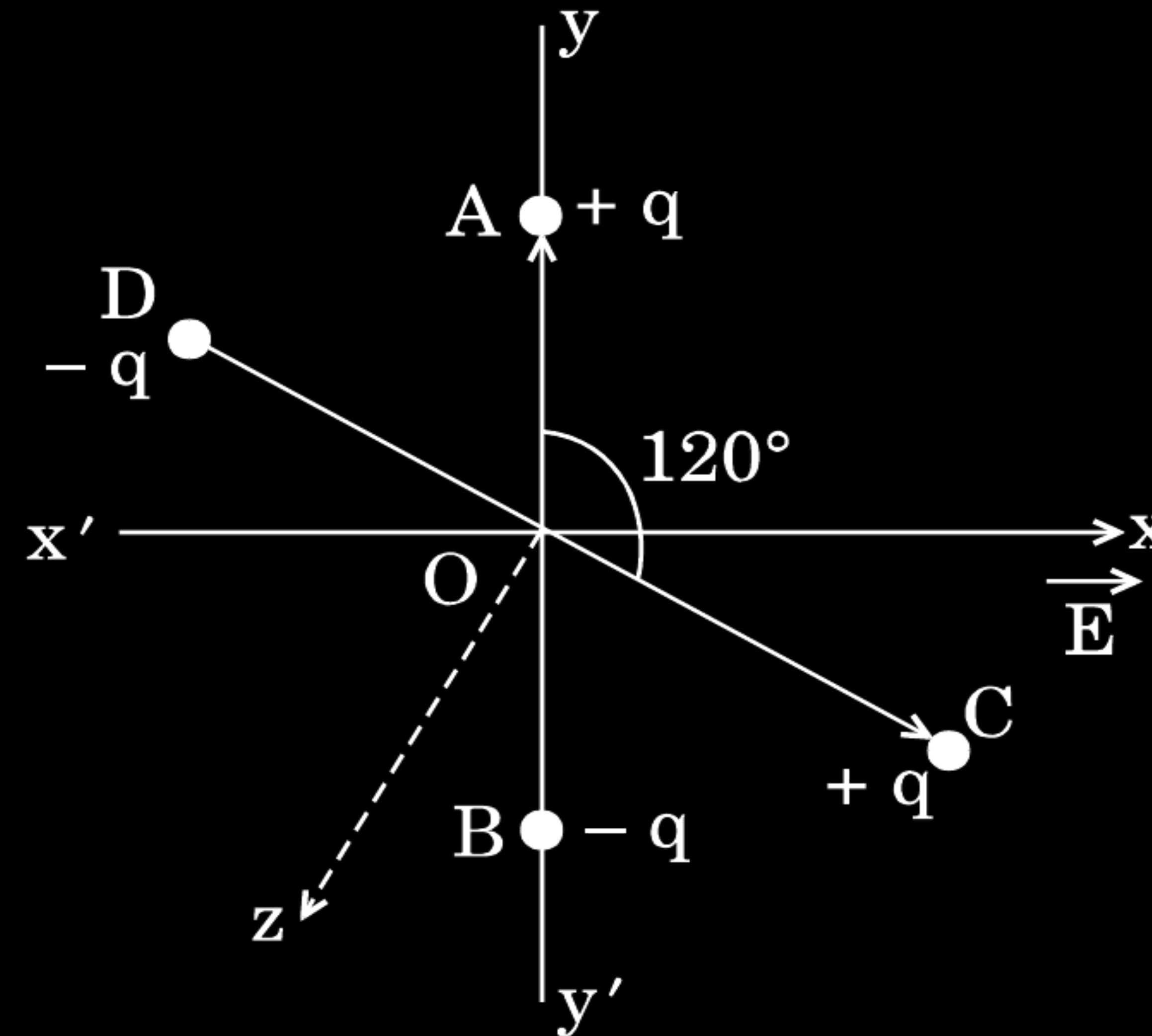
ii) For the torque to be half the maximum,

$$\theta = 30^\circ = \frac{\pi}{6}$$

1/2

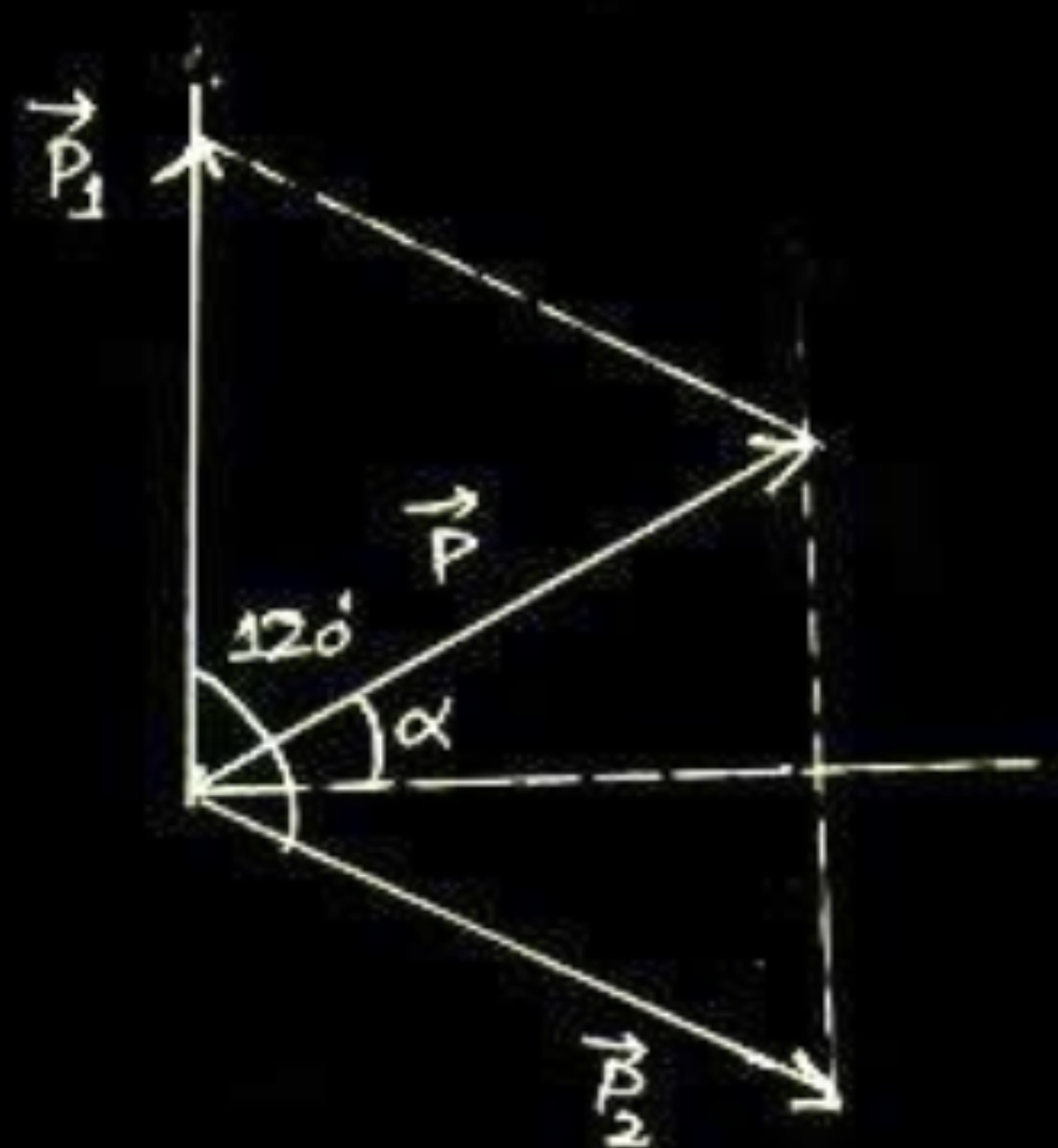
- 28.** Two small identical electric dipoles AB and CD, each of dipole moment $\vec{p}$ are kept at an angle of 120° to each other in an external electric field $\vec{E}$ pointing along the x-axis as shown in the figure. Find the
- dipole moment of the arrangement, and
 - magnitude and direction of the net torque acting on it.

3



(a) Magnitude & direction of net dipole moment	$1\frac{1}{2}$
(b) Magnitude & direction of net torque	$1\frac{1}{2}$

(a)



1/2

$$\begin{aligned} P &= (p_1^2 + p_2^2 + 2 p_1 p_2 \cos 120^\circ)^{1/2} \\ &= (2p^2 - p^2)^{1/2} \\ &= p \end{aligned}$$

1/2

Making 60° angle with $\vec{p}_1$ and $\alpha = 30^\circ$ (angle with X axis)

1/2

[$p_1 = p_2 = p$]

[Do not deduct 1/2 mark if diagram is not drawn but dipole moment and its direction are correctly worked out.]

If correct and complete vector diagram is drawn but dipole moment is not worked out then award 1 mark out of 1.5]

1/2

(b)

$$\begin{aligned} \vec{\tau} &= \vec{P} \times \vec{E} \\ \tau &= PE \sin 30^\circ \\ &= \frac{1}{2} pE \end{aligned}$$

1/2

1/2

Direction of $\vec{\tau}$ is into the plane of the paper or along -z direction.

Derive an expression for the torque acting on an electric dipole of dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$. Write the direction along which the torque acts.

OR

Derive an expression for the electric field at a point on the axis of an electric dipole of dipole moment $\vec{p}$. Also write its expression when the distance $r \gg$ the length 'a' of the dipole.

Diagram -	$\frac{1}{2}$
Expression for torque -	1
Direction of torque -	$\frac{1}{2}$

Expression for field at an axial point - $1\frac{1}{2}$

Field at large distance - $\frac{1}{2}$



Force on either charge $F = qE$

Magnitude of torque = Either of force $\times$ $\perp$ distance between them.

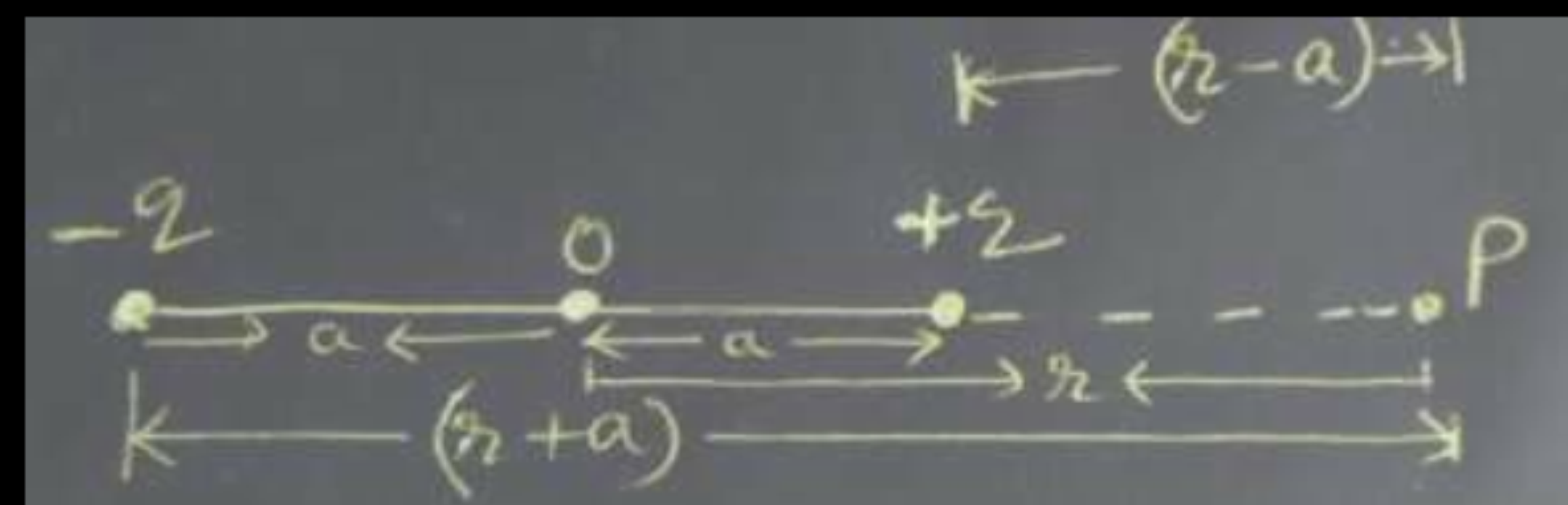
$$\tau = qE 2a \sin \theta$$

$$\tau = pE \sin \theta$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

Direction is normal to the paper coming out of it.

1/2



$$E_- = \frac{q}{4\pi\epsilon_0 (r-a)^2} \text{ along } (-)\vec{p}$$

$$E_+ = \frac{q}{4\pi\epsilon_0 (r+a)^2} \text{ along } \vec{p}$$

$\therefore$ Total field at P, $E = E_- - E_+$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{(r-a)^2} - \frac{1}{(r+a)^2} \right]$$

$$= \frac{q}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2}$$

For

$$r \gg a$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$$

1/2

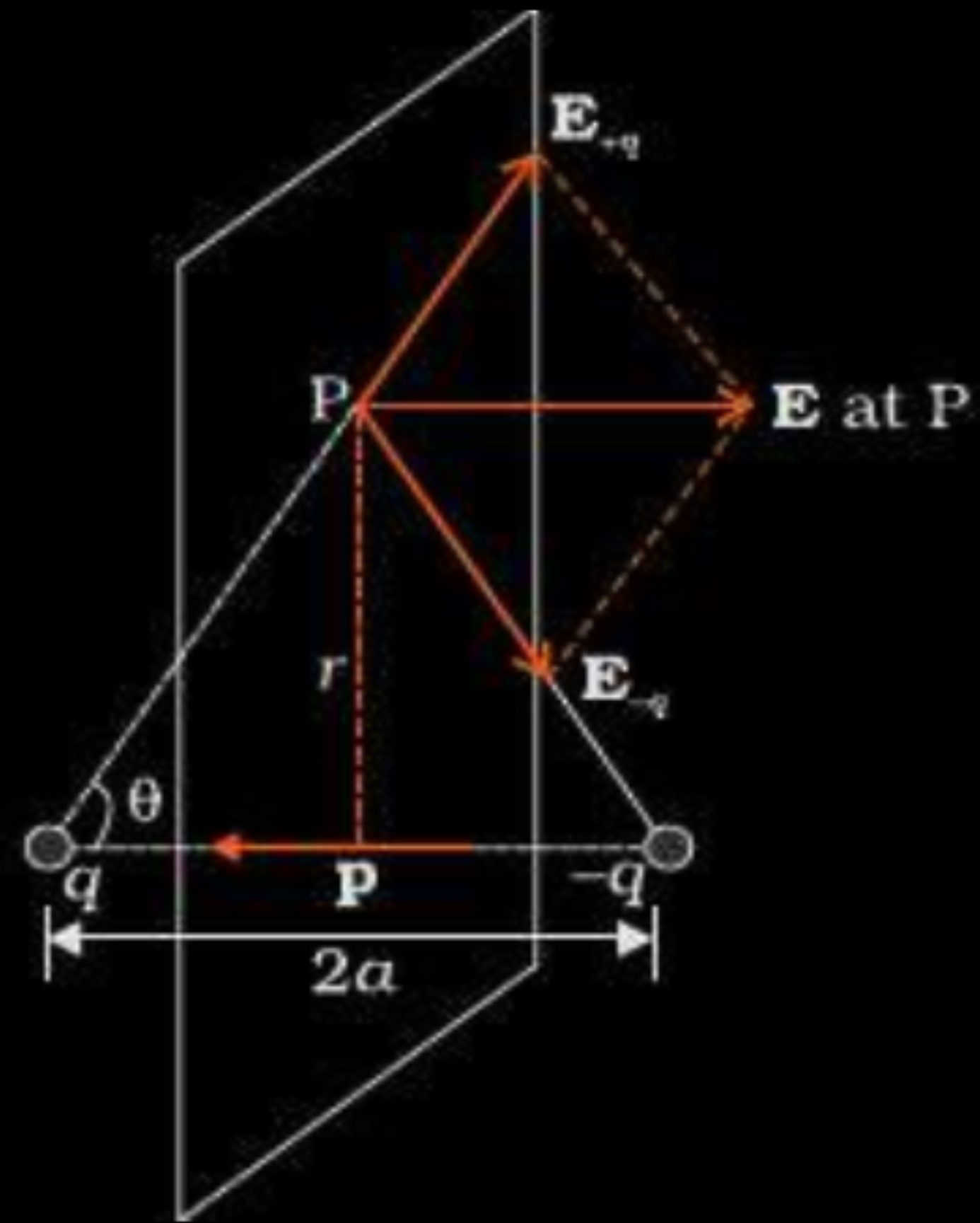
1/2

1/2

1/2

- (a) Derive an expression for the electric field at any point on the equatorial line of an electric dipole.

(a)



The magnitude of the electric fields due to the two charges $+q$ and $-q$ are

$$E_{+q} = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + a^2)}$$

$$E_{-q} = \frac{1}{4\pi\epsilon_0} \frac{q}{(r^2 + a^2)}$$

The components normal to the dipole axis cancel away and the components along the dipole axis add up

Hence total Electric field = $-(E_{+q} + E_{-q})\cos\theta \hat{p}$

$$E = -\frac{2qa}{4\pi\epsilon_0(r^2 + a^2)^{3/2}} \hat{p}$$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (a) Derive an expression for the electric field E due to a dipole of length ' $2a$ ' at a point distant r from the centre of the dipole on the axial line.
- (b) Draw a graph of E versus r for $r \gg a$.
- (c) If this dipole were kept in a uniform external electric field E_0 , diagrammatically represent the position of the dipole in stable and unstable equilibrium and write the expressions for the torque acting on the dipole in both the cases.

(a)



Electric field at P due to charge $(+q) = E_1 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2}$

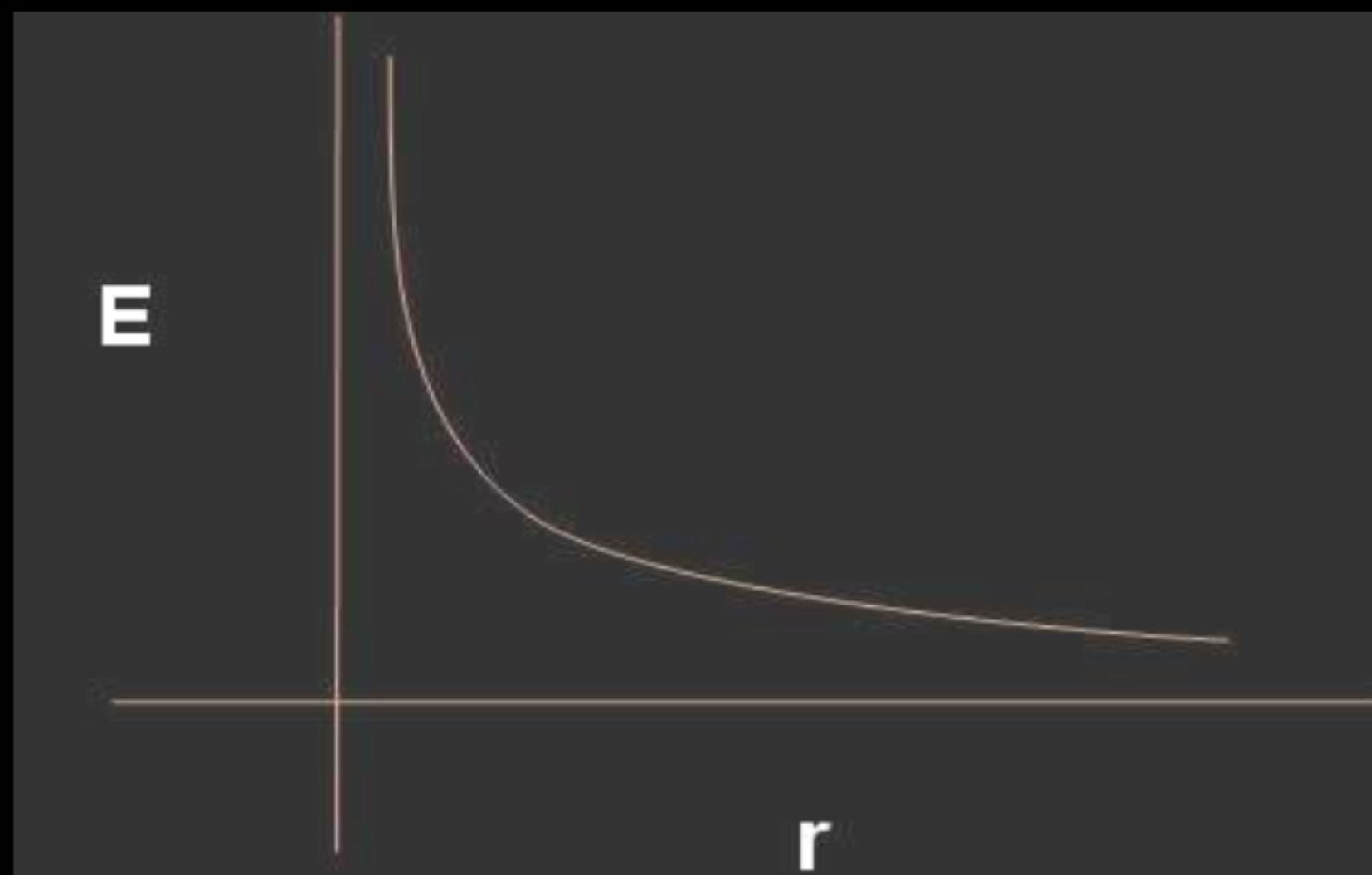
Electric field at P due to charge $(-q) = E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2}$

$$\text{Net electric Field at P} = E_1 - E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{(r-a)^2} - \frac{1}{4\pi\epsilon_0} \frac{q}{(r+a)^2}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{2pr}{(r^2 - a^2)^2} \quad (p = q \cdot 2a)$$

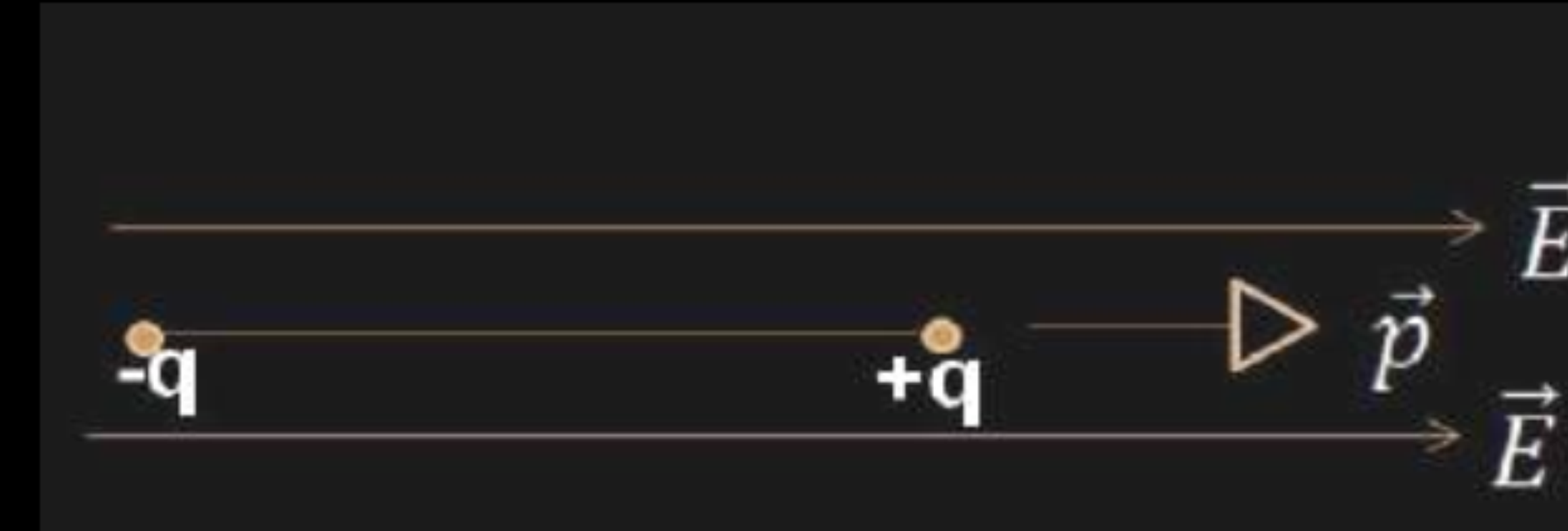
Its direction is parallel to $\vec{p}$.

(b)



(Note: Award $\frac{1}{2}$ mark if the student just writes: For short Dipole $= \frac{1}{4\pi\epsilon_0} \frac{2p}{r^3}$ without drawing the graph)

(c)



Stable equilibrium



Unstable equilibrium

(Note: Award $\frac{1}{2}$ mark only if the student does not draw the diagrams but just writes:

 $\frac{1}{2}$

(i) For stable Equilibrium: $\vec{p}$ is parallel to $\vec{E}$.

(ii) For unstable equilibrium: $\vec{p}$ is antiparallel to $\vec{E}$)

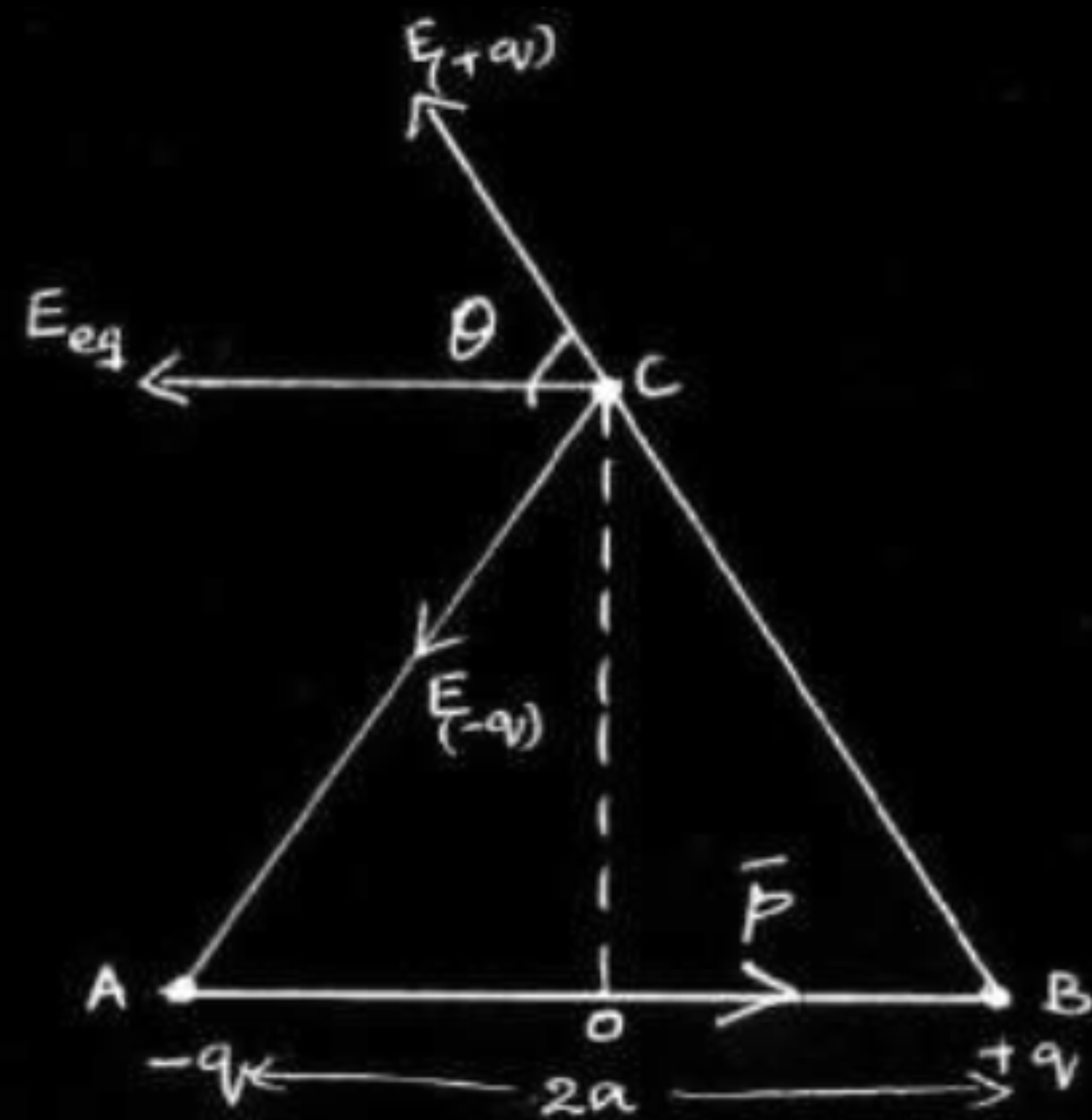
1

Torque = 0 for (i) as well as case (ii).

(Also accept, $\vec{\tau} = \vec{p} \times \vec{E}$ / $\tau = pE \sin \theta$)

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$

Derive an expression for the electric field intensity at a point on the equatorial line of an electric dipole of dipole moment $\vec{p}$ and length $2a$. What is the direction of this field ?



1

$$E_{+q} = Kq / (r^2 + a^2) \quad \text{and} \quad E_{-q} = Kq / (r^2 + a^2)$$

1/2

The two Electric fields have equal magnitudes and their directions are as shown in diagram

Components along dipole axis get added up while normal components cancel each other.

1/2

$$\therefore \vec{E} = -[E_{-q} + E_{+q}] \cos \theta \hat{r} \text{ so } E = - \frac{K2qa}{[r^2 + a^2]^{3/2}} \hat{r}$$

1/2

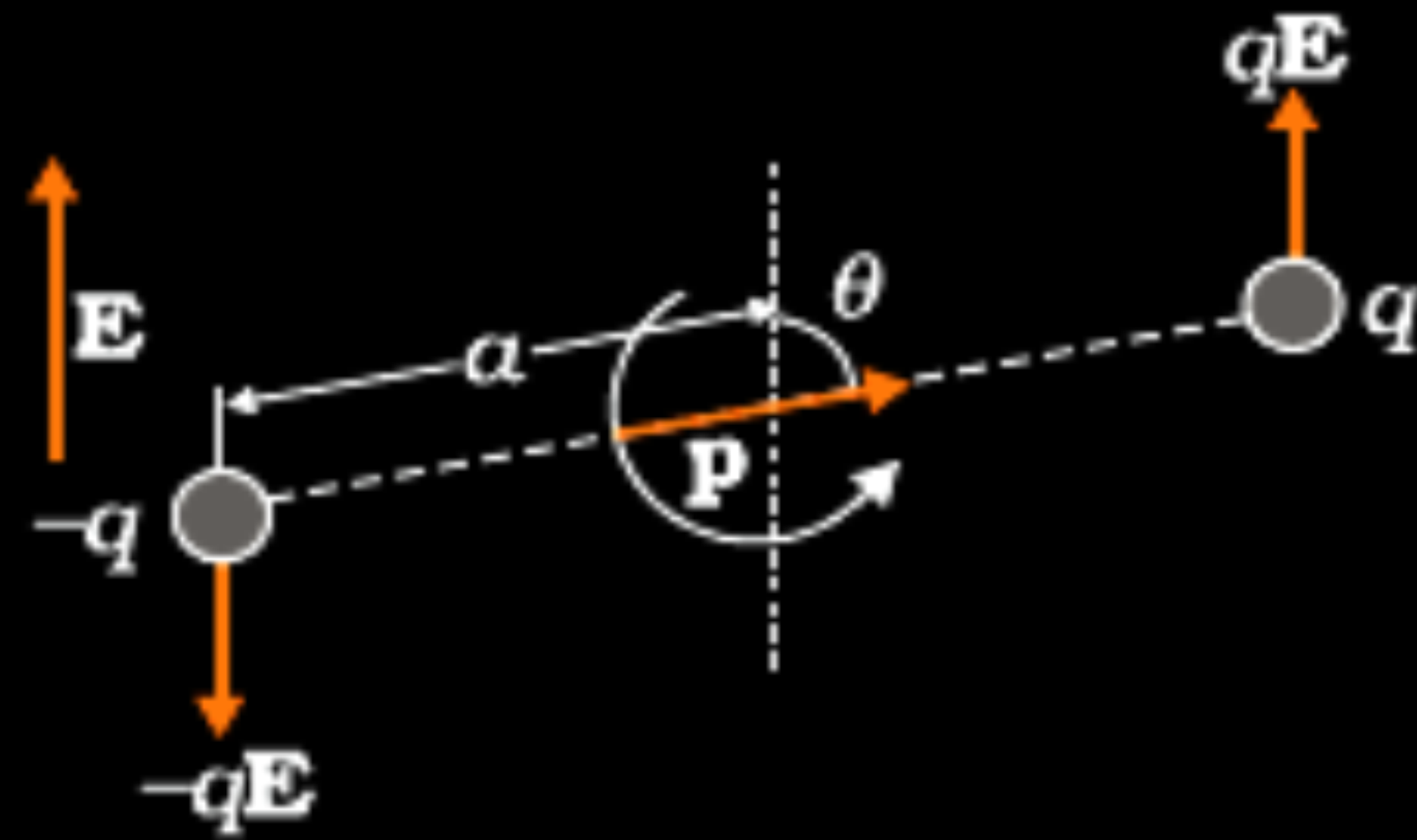
$$= \frac{kp}{[r^2 + a^2]^{3/2}} \quad (p = 2qa\hat{r}) = \frac{-1}{4\pi\epsilon_0} \frac{p}{[r^2 + a^2]^{3/2}}$$

$\therefore$ Direction of electric field is opposite to that of dipole moment.

1/2

- (a) Deduce the expression for the torque acting on a dipole of dipole moment $\vec{p}$ placed in a uniform electric field $\vec{E}$. Depict the direction of the torque. Express it in the vector form.

(a)



Magnitude of torque = magnitude of either force multiplied by the arm of the couple.

$$= qE \times 2a \sin \theta$$

$$= pE \sin \theta$$

Direction of torque is perpendicular to the plane containing $\vec{p}$ and $\vec{E}$.

Vector form $\vec{\tau} = \vec{p} \times \vec{E}$